

WEZU Energy Monetization Platform

Memorandum of Understanding Contract Document

Date: December 17, 2025

WEZU Energy Monetization Platform

Software Requirements Specification (SRS) Document

Version: 1.0

Date: December 17, 2025

Project: WEZU Battery Rental & Monetization Platform

Duration: 6 Months (Phase 1: 3 months, Phase 2: 3 months)

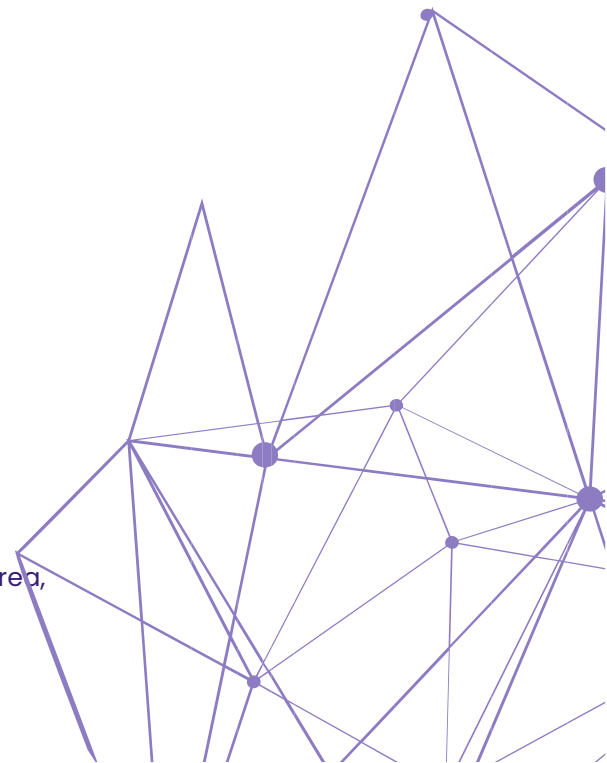
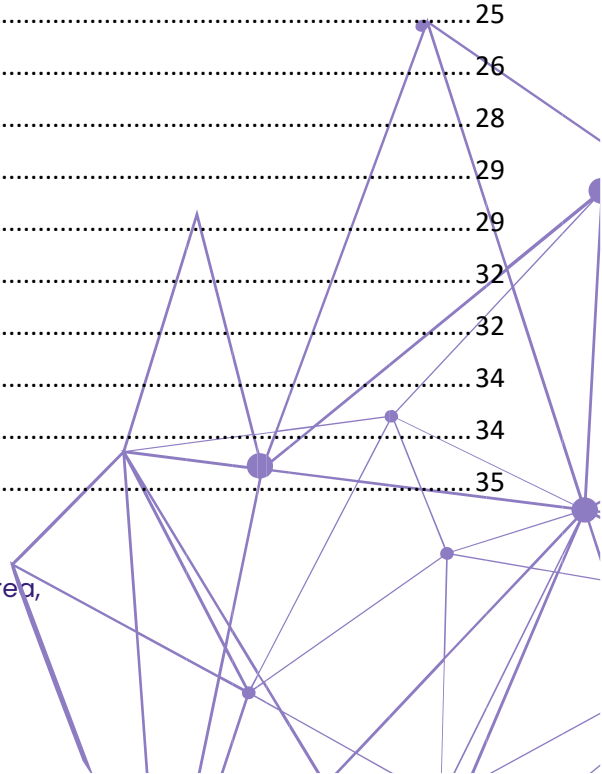


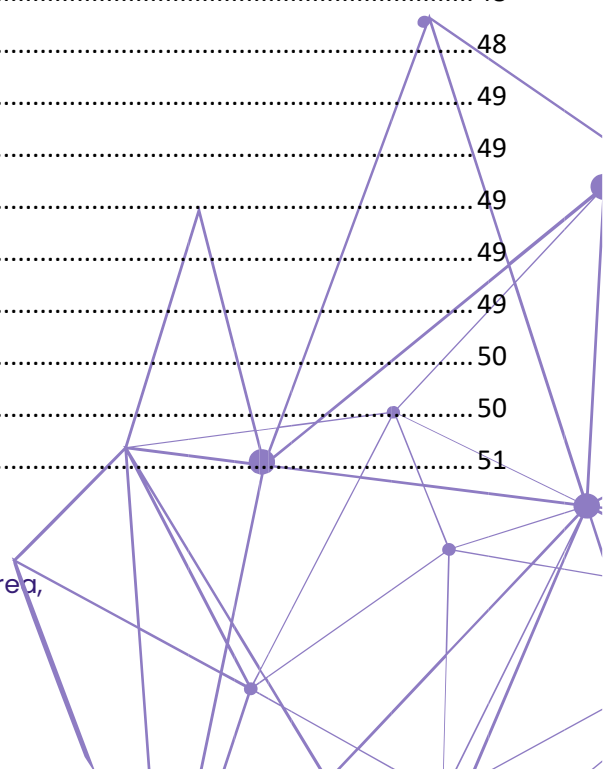
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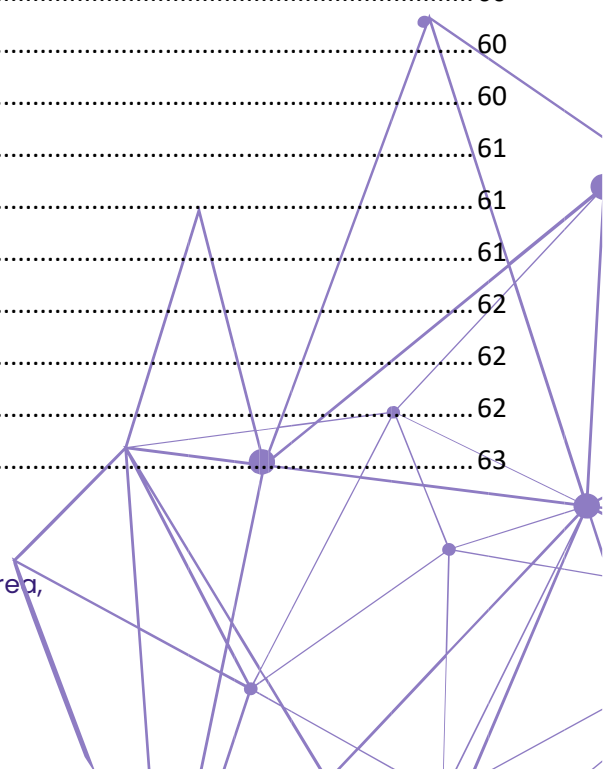
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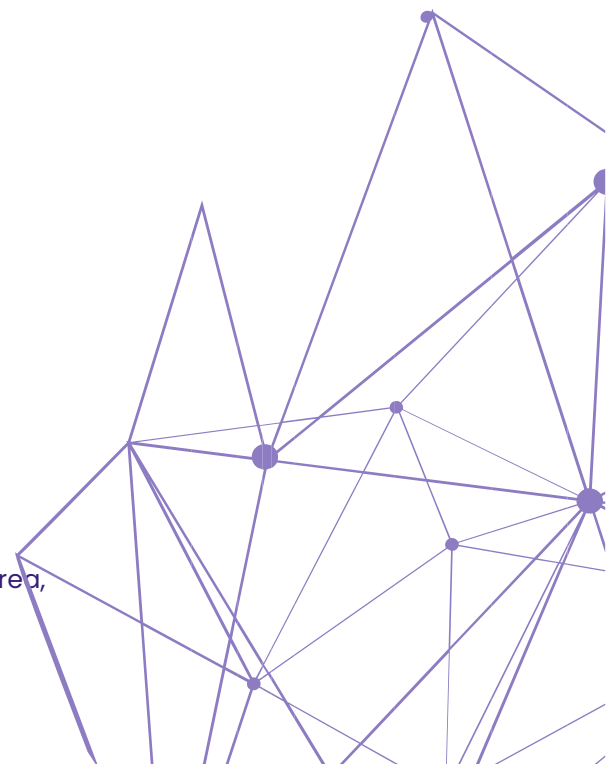
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Executive Summary

The WEZU Energy Monetization Platform is a comprehensive digital ecosystem designed to revolutionize battery rental, purchase, and logistics management. This platform supports four distinct user types (End-users, Administrators, Dealers, and Logistics Partners) through a multi-app architecture encompassing mobile applications, web dashboards, and specialized portals.

Key Platform Objectives:

- Enable seamless battery rental and purchasing across geographic regions
- Provide real-time GPS tracking and IoT device integration
- Process payments securely with multiple payment gateways
- Implement robust dealer KYC and fraud detection (95%+ accuracy)
- Scale to handle 10,000+ concurrent users with 99.9% uptime
- Deliver advanced AI/ML predictive analytics in Phase 2

Project Scope: Complete platform spanning battery discovery, rental management, payment processing, logistics coordination, and advanced analytics with fraud detection and revenue optimization capabilities.

2 Product Overview & Vision

2.1 Project Vision

To create a revolutionary battery-as-a-service platform that democratizes access to reliable energy solutions through an integrated digital ecosystem supporting rental, purchase, and logistics management with intelligent tracking and analytics.

2.2 Problem Statement

The battery rental industry lacks an integrated platform for:

- Seamless battery discovery and rental across multiple stations
- Real-time tracking and health monitoring of batteries
- Secure dealer onboarding with multi-layer fraud detection
- Intelligent inventory management across geographies
- Advanced analytics for demand forecasting and revenue optimization



2.3 Solution Overview

WEZU platform provides:

- Mobile app for customer battery rental and purchases
- Admin dashboard for platform management and analytics
- Dealer portal for station and inventory management
- Logistics platform for delivery coordination
- IoT integration for real-time tracking
- AI/ML models for fraud detection and predictive analytics

3 Stakeholder Analysis

3.1 User Personas & User Types

End-Users (Customer Mobile App Users)

- Role: Individual battery renters and purchasers
- Goals: Quick, convenient battery rental with transparent pricing
- Pain Points: Battery availability, tracking, late fees, payment convenience

Platform Administrators

- Role: Manage overall platform operations
- Goals: Monitor KPIs, manage users, ensure system health
- Pain Points: Real-time visibility, fraud detection, revenue tracking

Battery Dealers

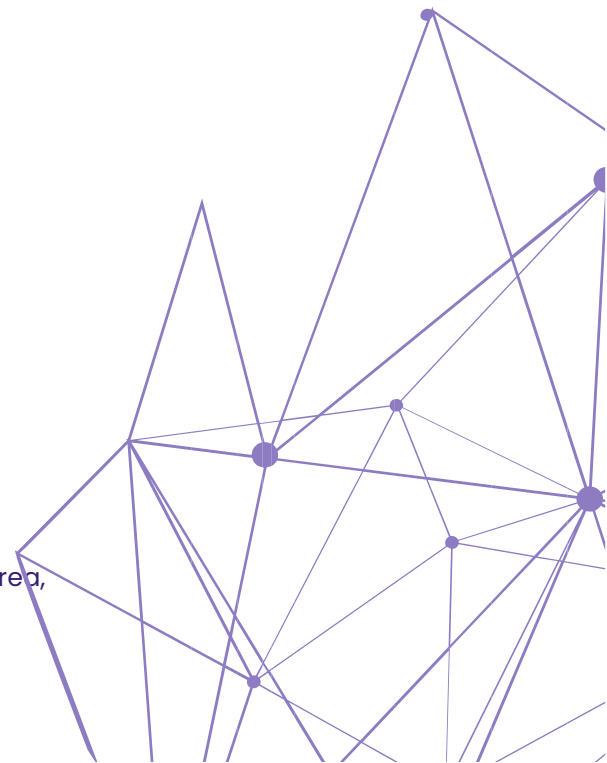
- Role: Manage charging stations and local inventory
- Goals: Maximize sales, track inventory, manage customers
- Pain Points: Onboarding complexity, inventory management, customer support

Logistics Partners

- Role: Manage battery delivery and reverse logistics
- Goals: Optimize routes, ensure timely delivery, reduce costs
- Pain Points: Route planning, real-time tracking, delivery proof

3.2 Stakeholder Requirements

- Transparent, secure transaction processing
- Real-time visibility into operations
- Compliance with local regulations (GST, KYC)
- Scalable infrastructure supporting growth



- Comprehensive reporting and analytics

4 Functional Requirements

4.1 Customer Mobile App (React Native) - Phase 1

4.1.1 Authentication & User Management

Requirement ID: FR-MOB-AUTH-001

Description: User registration with phone/email verification

Priority: Critical

Acceptance Criteria:

- Users can register using phone number or email
- OTP verification sent within 10 seconds
- OTP valid for 15 minutes
- Failed attempts logged for security

Requirement ID: FR-MOB-AUTH-002

Description: JWT-based secure login/logout

Priority: Critical

Acceptance Criteria:

- JWT tokens valid for 24 hours
- Secure token storage in device keychain
- Automatic logout after inactivity (30 minutes)
- Token refresh mechanism implemented

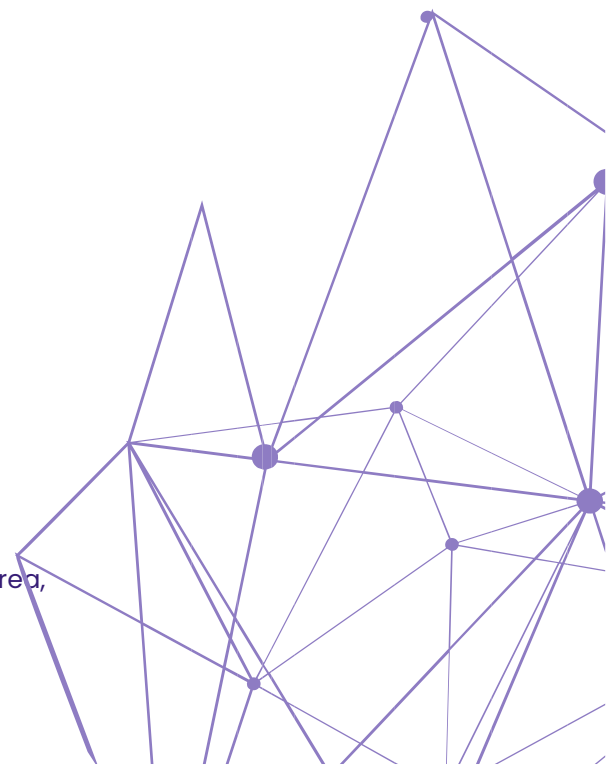
Requirement ID: FR-MOB-AUTH-003

Description: OAuth2 integration (Google & Apple Sign-in)

Priority: High

Acceptance Criteria:

- One-tap Google sign-in functional
- One-tap Apple sign-in functional
- Profile data auto-populated from OAuth provider
- User consent for data usage captured



Requirement ID: FR-MOB-AUTH-004**Description:** Complete user profile management**Priority:** High**Acceptance Criteria:**

- Users can update name, email, phone, address
- Profile picture upload with size limits (max 2MB)
- Address validation against postal database
- KYC verification status displayed

Requirement ID: FR-MOB-AUTH-005**Description:** KYC verification process**Priority:** Critical**Acceptance Criteria:**

- Aadhaar e-KYC integration for identity verification
- Video KYC with liveness detection (95%+ accuracy)
- PAN verification with NSDL database
- Address verification with utility bills
- Approval/rejection within 24 hours

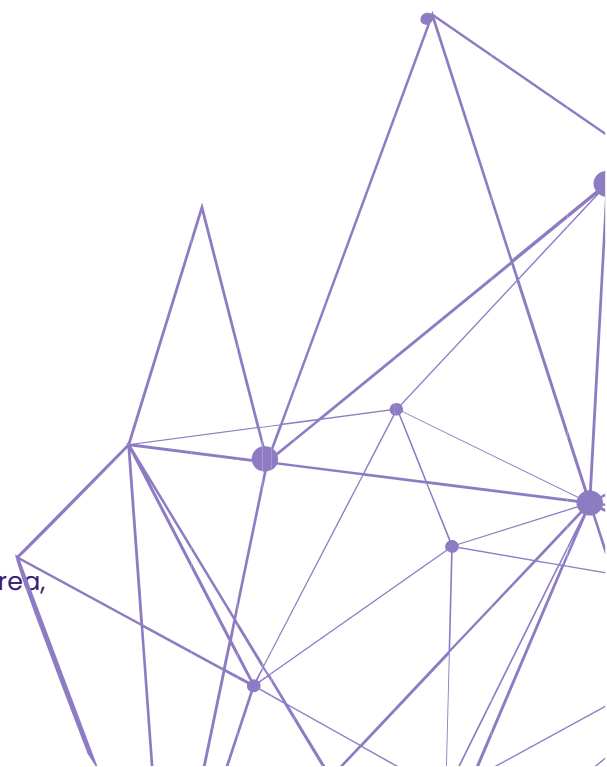
Requirement ID: FR-MOB-AUTH-006**Description:** Password reset & advanced security**Priority:** High**Acceptance Criteria:**

- Password reset link valid for 1 hour
- Two-factor authentication optional
- Biometric login support (fingerprint/face)
- Security question recovery mechanism

4.1.2 Battery Discovery & Location Services

Requirement ID: FR-MOB-DISC-001**Description:** Real-time Google Maps integration with live tracking**Priority:** Critical**Acceptance Criteria:**

- Map displays all nearby charging stations within 50km radius
- Real-time station status (operational/closed/maintenance)
- Live battery availability count shown on map
- Station details popup with opening hours

Requirement ID: FR-MOB-DISC-002**Description:** Intelligent nearby battery station locator

Priority: High**Acceptance Criteria:**

- Auto-detection of user location with permission
- Sorting by distance (nearest first)
- Sorting by availability (highest stock first)
- Sorting by user ratings (highest first)

Requirement ID: FR-MOB-DISC-003**Description:** Advanced filters for battery discovery**Priority:** High**Acceptance Criteria:**

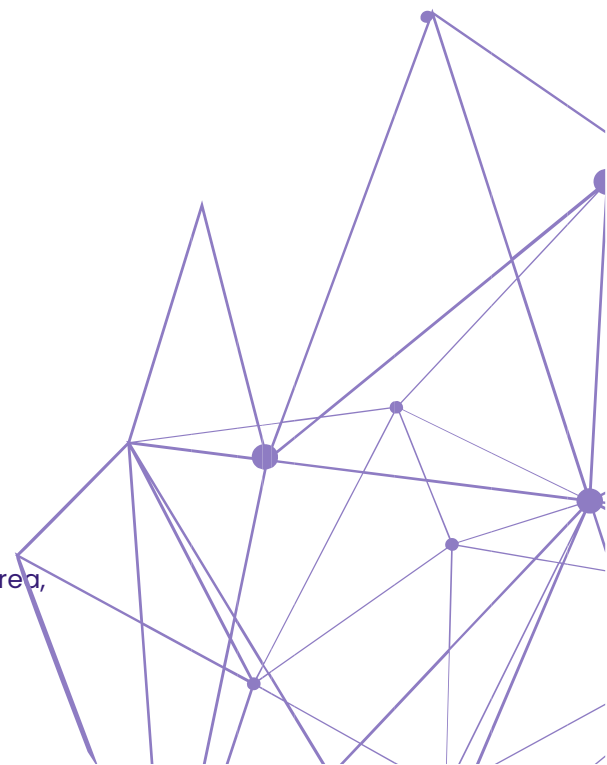
- Filter by location radius (1-50km)
- Filter by battery type and capacity
- Filter by price range
- Filter by user ratings (4+ stars)
- Filter by availability status

Requirement ID: FR-MOB-DISC-004**Description:** Detailed station information pages**Priority:** High**Acceptance Criteria:**

- Station name, address, contact information
- Gallery with 5+ photos of station
- User reviews and ratings (minimum 3 reviews)
- Operating hours and contact number
- Available battery inventory with details

Requirement ID: FR-MOB-DISC-005**Description:** Smart map clustering for visualization**Priority:** Medium**Acceptance Criteria:**

- Stations cluster when zoomed out (5+ clusters)
- Clusters show total station count
- Clusters expand when clicked
- Smooth animation transitions

Requirement ID: FR-MOB-DISC-006**Description:** GPS-based automatic location detection**Priority:** High**Acceptance Criteria:**

- Location permission request on first launch
- Continuous GPS accuracy within 10 meters
- Location updates every 5 seconds during rental
- Graceful handling when GPS unavailable

4.1.3 Battery Rental Flow

Requirement ID: FR-MOB-RENT-001

Description: Browse available batteries at selected station

Priority: Critical

Acceptance Criteria:

- Display list of all available batteries
- Show 5+ battery details per item (capacity, health %, age, type)
- Display images for each battery
- Show rental price and daily rate
- Sort by price, capacity, health percentage

Requirement ID: FR-MOB-RENT-002

Description: Detailed battery specifications display

Priority: High

Acceptance Criteria:

- Battery capacity in mAh (e.g., 5000mAh)
- Health percentage (0-100%)
- Age/cycle count information
- Battery type/model
- Warranty details
- Certification details

Requirement ID: FR-MOB-RENT-003

Description: QR code scanning for instant battery verification

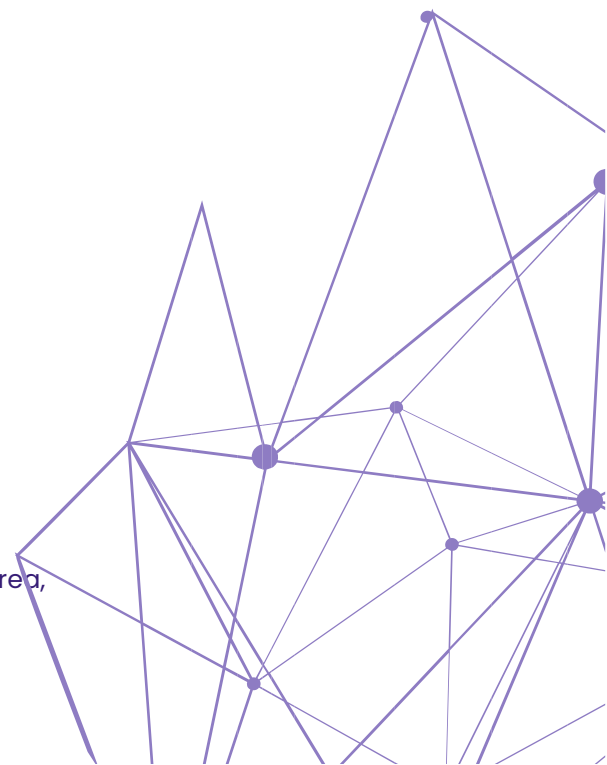
Priority: High

Acceptance Criteria:

- Camera permission request
- QR scan initiates battery details display
- Scan result shows battery serial number
- Verify battery matches selected item
- Time limit: scan completed within 5 seconds

Requirement ID: FR-MOB-RENT-004

Description: One-tap quick rental booking



Priority: Critical**Acceptance Criteria:**

- Single button to initiate rental
- Confirm rental duration (1-30 days)
- Review pricing breakdown
- Accept terms & conditions
- Rental confirmed within 2 seconds

Requirement ID: FR-MOB-RENT-005**Description:** Real-time battery monitoring**Priority:** High**Acceptance Criteria:**

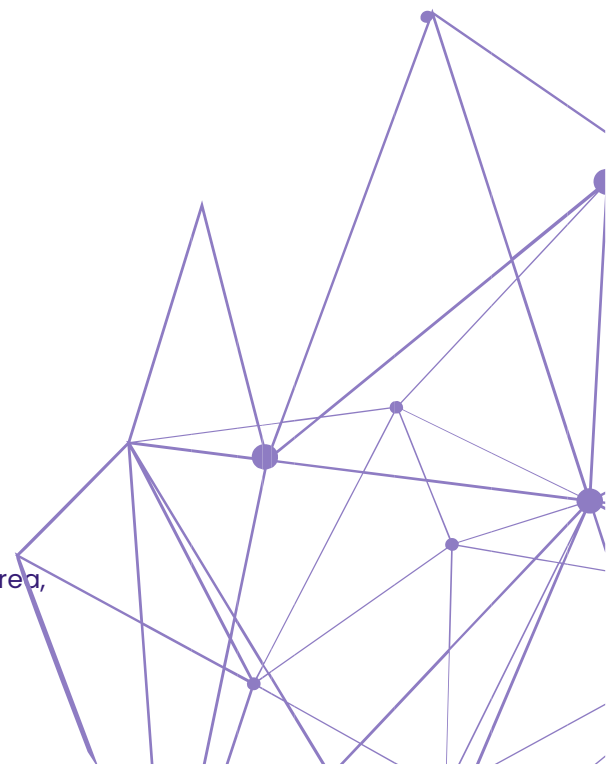
- Display current charge percentage (0-100%)
- Display battery health percentage (0-100%)
- Display battery temperature (°C)
- Display voltage (V)
- Update every 10 seconds via IoT device

Requirement ID: FR-MOB-RENT-006**Description:** Clear rental terms & pricing display**Priority:** High**Acceptance Criteria:**

- Daily rental rate clearly displayed
- Total cost calculation shown
- Damage deposit amount displayed
- Late fee structure transparent
- Discount/promo applied clearly

Requirement ID: FR-MOB-RENT-007**Description:** Instant booking confirmation with receipt**Priority:** High**Acceptance Criteria:**

- Confirmation page displayed immediately after booking
- Receipt generated with rental details
- Receipt downloadable as PDF
- Receipt sent via email/SMS
- Rental ID provided for reference



4.1.4 Battery Purchase Options

Requirement ID: FR-MOB-PURCH-001

Description: Comprehensive in-app battery purchase catalog

Priority: High

Acceptance Criteria:

- Catalog displays 50+ battery SKUs
- Search functionality with autocomplete
- Category browsing (by type, capacity, brand)
- Filter by price range, capacity, reviews
- Sort by popularity, price, rating

Requirement ID: FR-MOB-PURCH-002

Description: Multiple battery SKU options

Priority: High

Acceptance Criteria:

- Display 5+ capacity options (2000-10000mAh)
- Display 3+ brand options
- Display different color options
- Display warranty tiers (1yr, 2yr, 3yr)
- Inventory count per SKU

Requirement ID: FR-MOB-PURCH-003

Description: Transparent pricing & warranty information

Priority: High

Acceptance Criteria:

- Price per unit clearly displayed
- Bulk purchase discounts shown
- Warranty period in months displayed
- Money-back guarantee information
- Warranty claim process documented

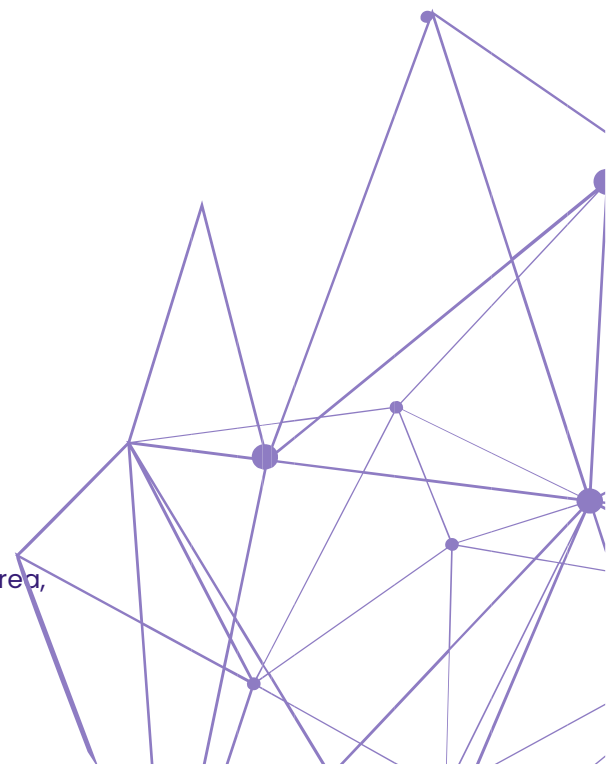
Requirement ID: FR-MOB-PURCH-004

Description: Secure Razorpay payment gateway

Priority: Critical

Acceptance Criteria:

- PCI-DSS Level 1 compliance
- All payment methods supported (UPI, cards, wallets)
- Payment confirmed within 2 seconds
- Failed payment clear error messaging



- Automatic refund for failed transactions

Requirement ID: FR-MOB-PURCH-005

Description: Instant invoice generation & delivery

Priority: High

Acceptance Criteria:

- Invoice generated within 1 second of payment
- PDF download option available
- Invoice sent via email automatically
- Invoice sent via SMS with order number
- Invoice includes GST/tax details

Requirement ID: FR-MOB-PURCH-006

Description: Real-time order tracking

Priority: High

Acceptance Criteria:

- Order status updated in real-time
- Tracking number provided
- Expected delivery date shown
- Delivery partner contact information
- Delivery proof with photo capture

4.1.5 Active Rental Management

Requirement ID: FR-MOB-ACTIVE-001

Description: Live GPS tracking of rented batteries

Priority: Critical

Acceptance Criteria:

- Real-time battery location shown on map
- Battery marker updated every 5 seconds
- Battery location accurate within 10 meters
- Geo-fence alerts when battery moves outside 5km radius
- Location history available for past 30 days

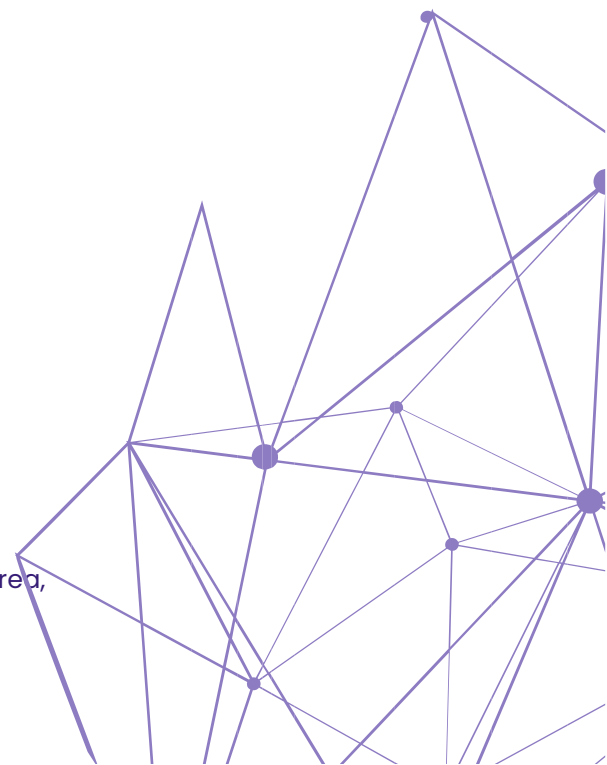
Requirement ID: FR-MOB-ACTIVE-002

Description: Real-time battery health monitoring

Priority: High

Acceptance Criteria:

- Voltage display with min/max values
- Temperature monitoring (normal/warning/critical)
- Charge percentage displayed real-time



- Health percentage with degradation trend
- Health alerts when below 80%

Requirement ID: FR-MOB-ACTIVE-003

Description: Digital countdown timer for rental

Priority: High

Acceptance Criteria:

- Timer shows remaining rental hours/days
- Countdown updates every minute
- Visual warning 24 hours before expiry
- Push notification 12 hours before expiry
- Final reminder 1 hour before expiry

Requirement ID: FR-MOB-ACTIVE-004

Description: Smart battery swap notifications

Priority: High

Acceptance Criteria:

- Notifications sent 24 hours before rental expiry
- Notifications sent 12 hours before expiry
- Notifications sent 1 hour before expiry
- Notification includes nearby swap stations
- One-tap to initiate swap process

Requirement ID: FR-MOB-ACTIVE-005

Description: One-tap battery swap request

Priority: High

Acceptance Criteria:

- Swap request initiated with single button
- Confirm new rental duration
- Select swap location from 3 nearest options
- Estimate time to reach station
- Confirm swap request

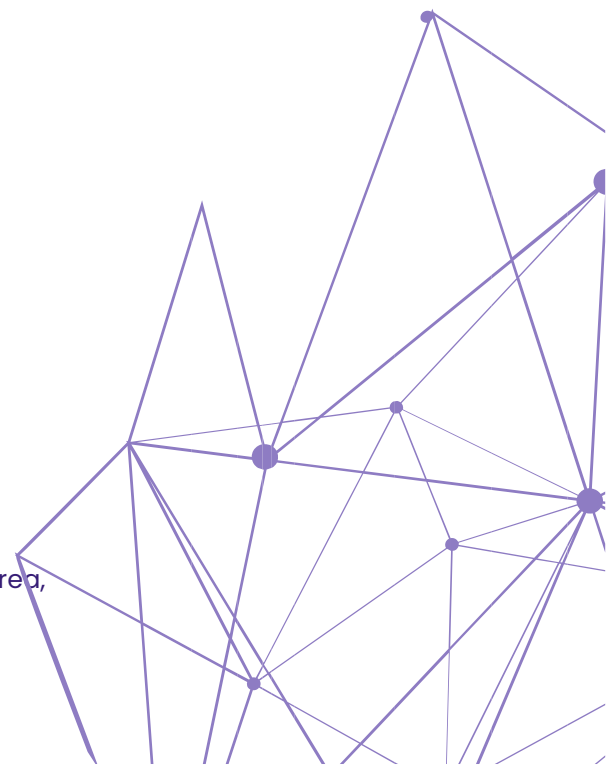
Requirement ID: FR-MOB-ACTIVE-006

Description: Intelligent return/swap station suggestions

Priority: Medium

Acceptance Criteria:

- Suggest 3 nearest stations to current location
- Show stock availability at each station
- Show distance and travel time



- Show hours and capacity
- Allow filtering by battery type needed

Requirement ID: FR-MOB-ACTIVE-007

Description: Automated late fee calculator

Priority: High

Acceptance Criteria:

- Calculate fees for each hour late
- Display cumulative late fees in real-time
- Show daily cap on late fees (if applicable)
- Provide payment option to clear dues
- Generate invoice for late fees

4.1.6 Payment & Transaction Management

Requirement ID: FR-MOB-PAY-001

Description: Multiple payment methods support

Priority: Critical

Acceptance Criteria:

- UPI payment integration (Google Pay, PhonePe, etc.)
- Debit card payment support
- Credit card payment support
- Digital wallet integration (Paytm, Amazon Pay)
- Net banking option available

Requirement ID: FR-MOB-PAY-002

Description: Secure PCI-DSS compliant Razorpay

Priority: Critical

Acceptance Criteria:

- PCI-DSS Level 1 compliance verified
- Payment encryption with TLS 1.2+
- Tokenization of card details (no storage)
- 3D Secure authentication for cards
- Fraud detection enabled

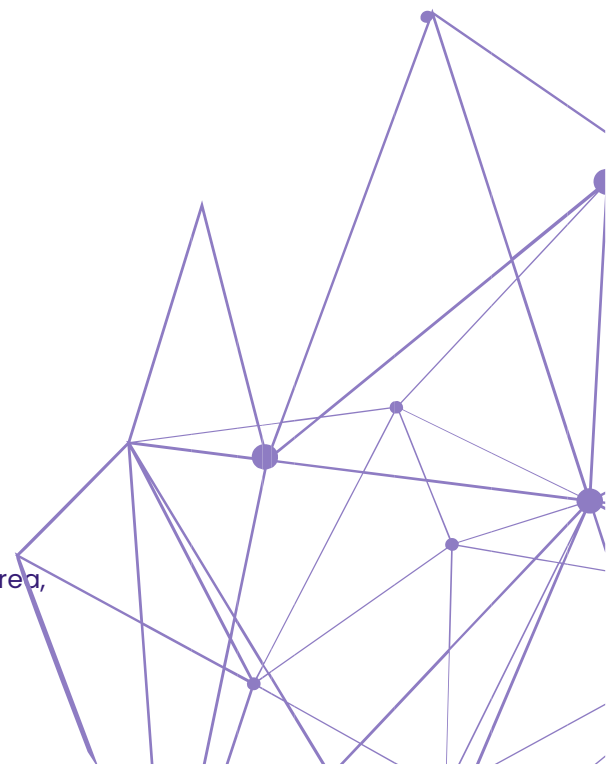
Requirement ID: FR-MOB-PAY-003

Description: Complete transaction history

Priority: High

Acceptance Criteria:

- All transactions listed with dates
- Transaction amount clearly displayed



- Payment method shown
- Transaction status (success/failed/refunded)
- Searchable by date range or amount

Requirement ID: FR-MOB-PAY-004

Description: Receipt download and email

Priority: High

Acceptance Criteria:

- PDF receipt download option
- Receipt sent via email automatically
- Receipt sent via SMS option
- Receipt includes transaction ID
- Receipt includes GST/tax breakdown

Requirement ID: FR-MOB-PAY-005

Description: Digital wallet and prepaid account

Priority: Medium

Acceptance Criteria:

- User can load wallet with ₹100-₹10,000
- Wallet balance displayed in profile
- Wallet used automatically for transactions
- Cashback earned on wallet transactions (1-5%)
- Wallet withdrawal option available

Requirement ID: FR-MOB-PAY-006

Description: Transparent refund tracking

Priority: High

Acceptance Criteria:

- Refund status visible in transaction history
- Refund amount clearly shown
- Refund processing time displayed
- Refund timeline: 3-5 business days
- Automatic notification when refund processed

4.1.7 Support & Notifications

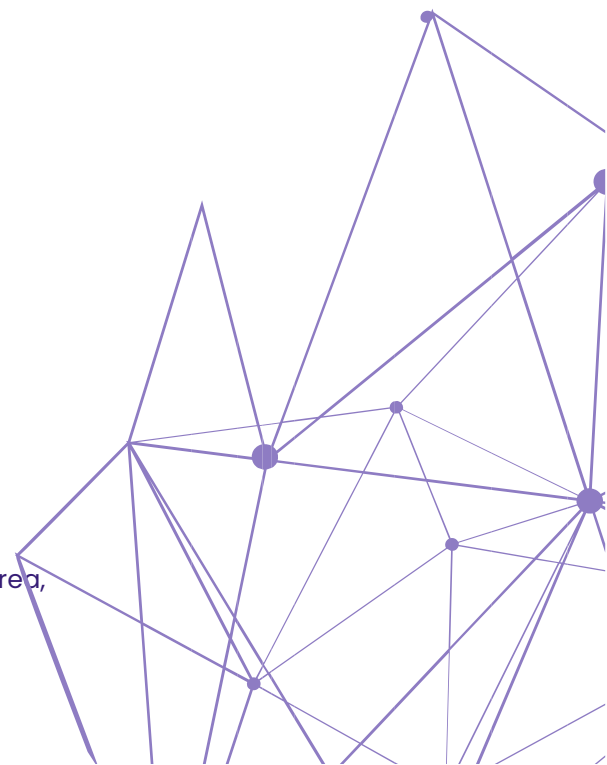
Requirement ID: FR-MOB-SUP-001

Description: Real-time push notifications

Priority: High

Acceptance Criteria:

- Push notifications for rental confirmations



- Notifications for deals and offers
- Notifications for system alerts
- In-app notification badge with count
- Notification preferences customizable

Requirement ID: FR-MOB-SUP-002**Description:** In-app messaging system**Priority:** High**Acceptance Criteria:**

- Chat interface with support team
- Automated response for common queries
- Message history searchable
- Estimated response time shown (2 hours)
- File/image attachment support

Requirement ID: FR-MOB-SUP-003**Description:** Live chat with automated responses**Priority:** High**Acceptance Criteria:**

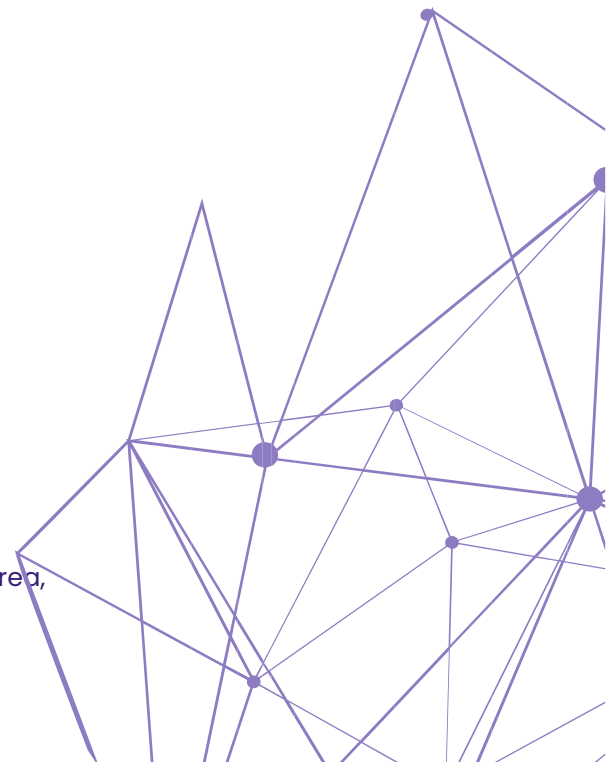
- Live chat icon visible in app
- Chat support available 24/7 (automated or live)
- Initial response within 2 minutes
- Average resolution time within 4 hours
- Chat transcript sent via email

Requirement ID: FR-MOB-SUP-004**Description:** Comprehensive FAQ & guides**Priority:** Medium**Acceptance Criteria:**

- FAQ section with 50+ common questions
- Searchable by keyword
- Video guides for key processes (rent, swap, purchase)
- Troubleshooting guides
- Categorized by topic (billing, technical, etc.)

Requirement ID: FR-MOB-SUP-005**Description:** Smart battery alert notifications**Priority:** Medium**Acceptance Criteria:**

- Alert when battery charge below 10%



- Alert when battery health below 80%
- Alert when battery temperature abnormal
- Alert for maintenance due dates
- Maintenance scheduling from notification

Requirement ID: FR-MOB-SUP-006

Description: Personalized promotional notifications

Priority: Medium

Acceptance Criteria:

- Offers based on user rental history
- Personalized discount codes sent
- Seasonal promotions targeted
- Birthday offers automatic
- Frequency capped (max 3 per week)

4.1.8 User Analytics & Reports

Requirement ID: FR-MOB-ANAL-001

Description: Complete rental history with timestamps

Priority: High

Acceptance Criteria:

- All rentals listed chronologically (newest first)
- Rental start and end dates shown
- Battery serial number listed
- Station name and location shown
- Rental duration and cost displayed

Requirement ID: FR-MOB-ANAL-002

Description: Personal cost analytics

Priority: Medium

Acceptance Criteria:

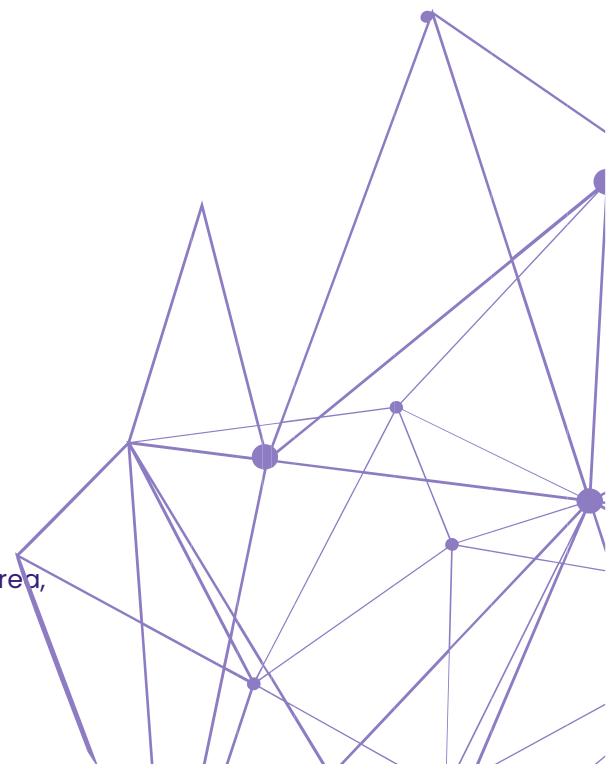
- Total spending displayed (monthly/yearly)
- Breakdown by rental vs purchase
- Cost trend chart (3 months, 6 months, 1 year)
- Average monthly spending calculated
- Comparison with previous period

Requirement ID: FR-MOB-ANAL-003

Description: Battery usage statistics

Priority: Medium

Acceptance Criteria:



- Total batteries rented displayed
- Average rental duration shown
- Most rented battery type
- Usage pattern analysis (by hour/day/week)
- Carbon saved calculation

Requirement ID: FR-MOB-ANAL-004

Description: Favorite stations & batteries

Priority: Low

Acceptance Criteria:

- Mark stations as favorite (heart icon)
- Favorite list accessible from home screen
- Show distance from current location
- Sort by rating/distance/availability
- Quick access to favorite stations

Requirement ID: FR-MOB-ANAL-005

Description: Personal dashboard with stats

Priority: High

Acceptance Criteria:

- Dashboard shows total spent this month
- Dashboard shows active rentals count
- Dashboard shows total rentals lifetime
- Dashboard shows membership tier/status
- Dashboard shows quick action buttons

4.2 Admin Dashboard (React Web) - Phase 1

4.2.1 Dashboard & Analytics

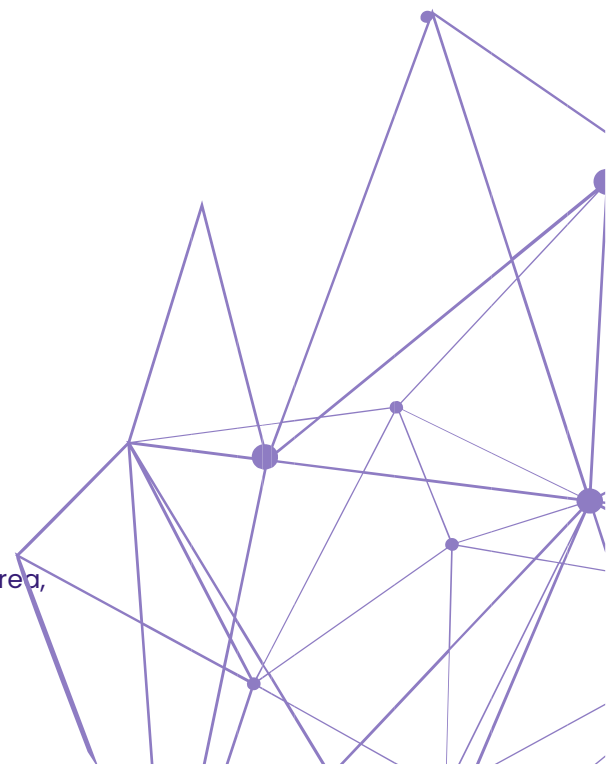
Requirement ID: FR-ADMIN-DASH-001

Description: Real-time platform metrics & KPIs

Priority: Critical

Acceptance Criteria:

- Dashboard loads within 2 seconds
- Real-time data refresh every 10 seconds
- Responsive design for desktop/tablet
- Dark/light theme options
- Customizable widget layout



Requirement ID: FR-ADMIN-DASH-002**Description:** Total rentals, revenue, active users tracking**Priority:** Critical**Acceptance Criteria:**

- Total rentals today, this week, this month displayed
- Total revenue with currency conversion
- Active users in last 24 hours shown
- Month-over-month comparison percentages
- Trend indicators (up/down arrows)

Requirement ID: FR-ADMIN-DASH-003**Description:** Daily, weekly, monthly trend analysis**Priority:** High**Acceptance Criteria:**

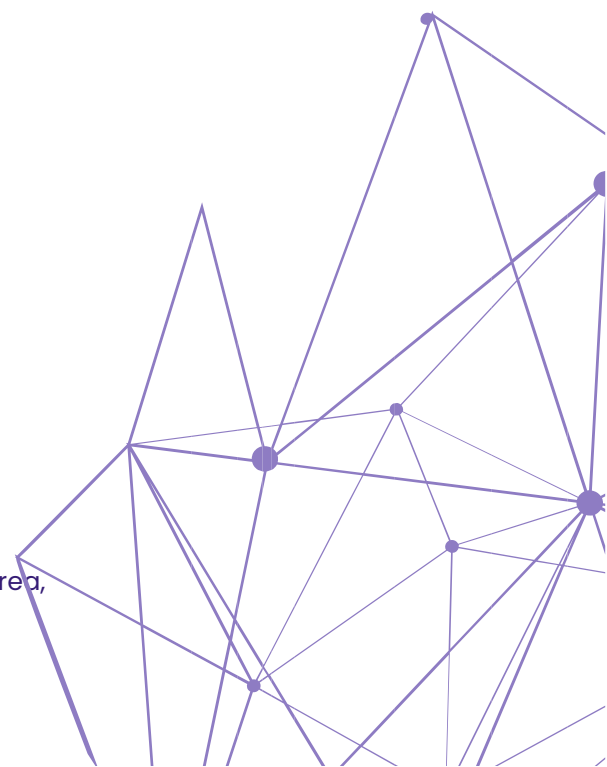
- Line chart showing daily trends (last 30 days)
- Comparison period selectable (last month, 3 months, 1 year)
- CSV export option for charts
- Hoverable tooltips with exact values
- Multi-metric chart overlays

Requirement ID: FR-ADMIN-DASH-004**Description:** Battery health distribution charts**Priority:** High**Acceptance Criteria:**

- Pie chart showing batteries by health (0-25%, 25-50%, etc.)
- Total batteries tracked shown
- Batteries needing maintenance highlighted
- Alert if >20% batteries below 80% health
- Maintenance schedule recommendation

Requirement ID: FR-ADMIN-DASH-005**Description:** Revenue analytics by station and type**Priority:** High**Acceptance Criteria:**

- Bar chart showing revenue per station
- Filter by date range (1 day to 1 year)
- Revenue breakdown by battery type
- Commission tracking by dealer
- Profitability margins calculated



Requirement ID: FR-ADMIN-DASH-006**Description:** Advanced conversion funnel visualization**Priority:** Medium**Acceptance Criteria:**

- Funnel chart: app installs → registrations → first rental
- Each stage shows conversion percentage
- Drop-off analysis at each stage
- Comparison with previous period
- Drill-down into each stage

4.2.2 Fleet & Inventory Management

Requirement ID: FR-ADMIN-INV-001**Description:** Complete battery inventory tracking**Priority:** Critical**Acceptance Criteria:**

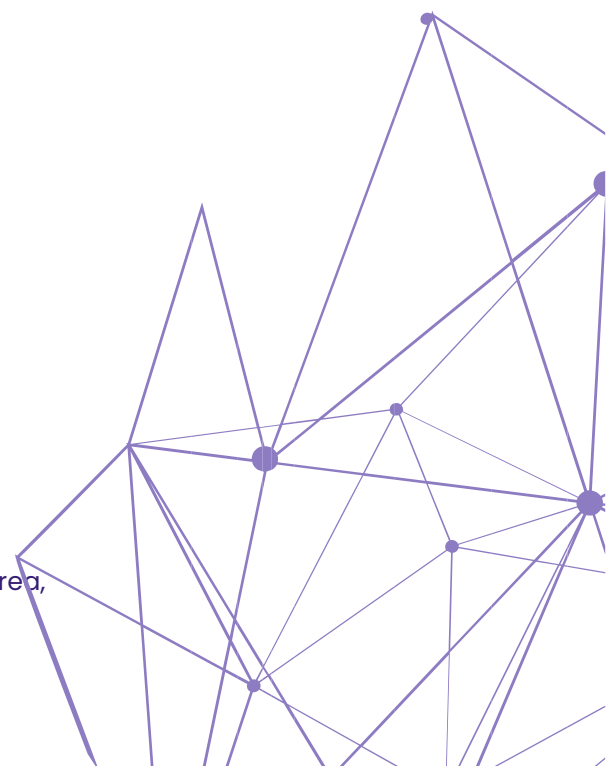
- All batteries listed with serial numbers
- Current location per battery shown
- Availability status (available/rented/maintenance)
- Health percentage displayed
- Last updated timestamp

Requirement ID: FR-ADMIN-INV-002**Description:** Real-time stock levels & availability**Priority:** Critical**Acceptance Criteria:**

- Stock count per station displayed
- Low stock alerts (when <10% capacity)
- Available units vs total units shown
- Utilization percentage calculated
- Reorder point configurable per station

Requirement ID: FR-ADMIN-INV-003**Description:** Inventory forecasting & reorder alerts**Priority:** High**Acceptance Criteria:**

- Demand forecast for next 30 days
- Recommended reorder quantity shown
- Automatic alerts when stock falls below threshold
- SMS/email alerts to depot manager



- Reorder request form pre-populated

Requirement ID: FR-ADMIN-INV-004

Description: Battery serial number management

Priority: High

Acceptance Criteria:

- Each battery assigned unique serial number
- Serial number barcode generation
- QR code for mobile scanning
- Serial number searchable
- RFID tag mapping

Requirement ID: FR-ADMIN-INV-005

Description: Inventory audit trails & history

Priority: High

Acceptance Criteria:

- Track all inventory changes with timestamp
- User who made change logged
- Previous/new values recorded
- Reason for change captured
- Audit trail searchable and exportable

Requirement ID: FR-ADMIN-INV-006

Description: Bulk inventory import/export

Priority: Medium

Acceptance Criteria:

- Import batteries via CSV/Excel
- Export inventory to CSV/Excel
- Batch operations (update 100+ items)
- Template provided for import
- Validation errors shown clearly

4.2.3 Station Management

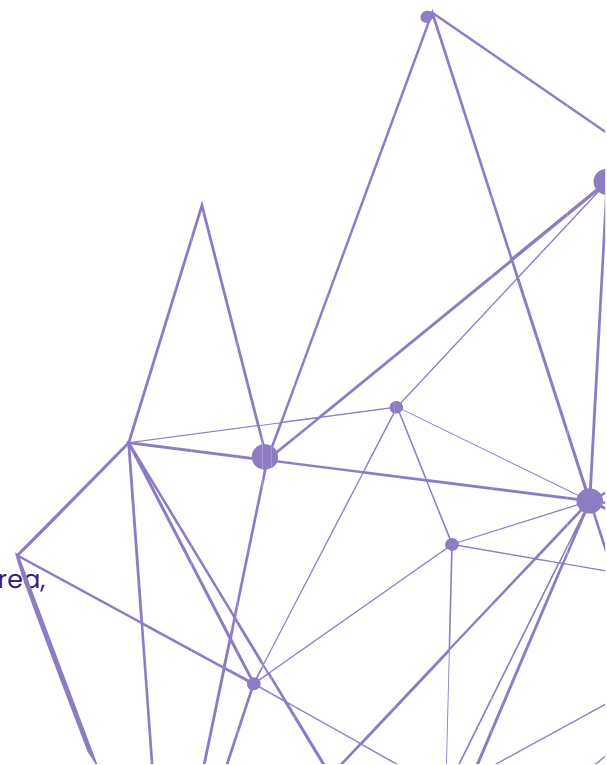
Requirement ID: FR-ADMIN-STN-001

Description: Create and manage charging stations

Priority: Critical

Acceptance Criteria:

- Admin can add new station with details
- Station name, address, coordinates required
- Contact information captured



- Operating hours configured
- Capacity (max batteries) set

Requirement ID: FR-ADMIN-STN-002

Description: Configure station capacity & specs

Priority: High

Acceptance Criteria:

- Maximum battery capacity per station
- Charger count and type
- Power requirements specified
- Temperature control availability
- Safety features documented

Requirement ID: FR-ADMIN-STN-003

Description: Real-time station status monitoring

Priority: High

Acceptance Criteria:

- Status: Operational, Closed, Maintenance, Error shown
- Green/yellow/red indicator lights
- Current active rentals displayed
- Available batteries count
- Last updated timestamp

Requirement ID: FR-ADMIN-STN-004

Description: Station performance metrics

Priority: High

Acceptance Criteria:

- Daily rentals count per station
- Average rental duration
- Revenue per station
- Customer satisfaction rating
- Utilization percentage

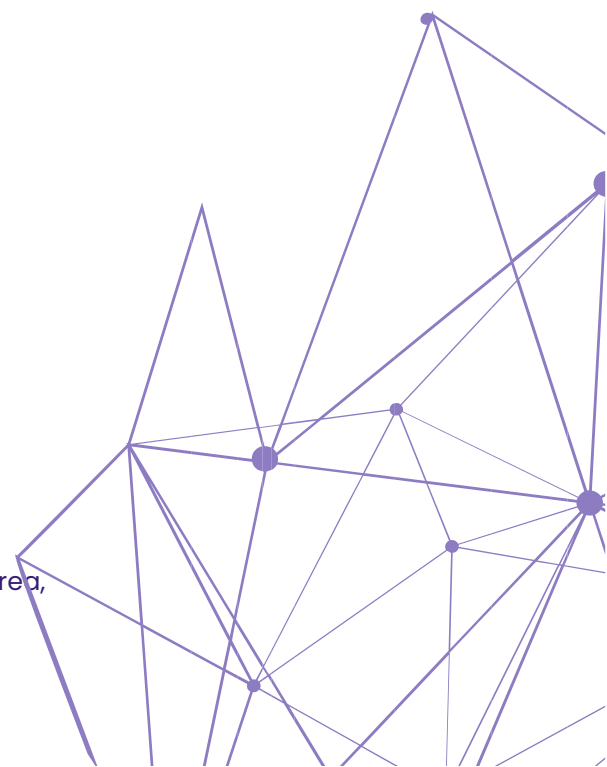
Requirement ID: FR-ADMIN-STN-005

Description: Automated maintenance scheduling

Priority: Medium

Acceptance Criteria:

- Maintenance calendar view
- Recurring maintenance schedule
- Maintenance history with dates



- Alerts for overdue maintenance
- Maintenance checklist template

Requirement ID: FR-ADMIN-STN-006

Description: Geo-location mapping of stations

Priority: High

Acceptance Criteria:

- Interactive map showing all stations
- Cluster view for many stations
- Click to see station details
- Export station map as image
- Heatmap showing high-demand areas

4.2.4 User & Account Management

Requirement ID: FR-ADMIN-USER-001

Description: User account creation & management

Priority: High

Acceptance Criteria:

- Admin can manually create user accounts
- Pre-populate with user details
- Set initial password or send invite link
- User profile accessible
- Account modification available

Requirement ID: FR-ADMIN-USER-002

Description: Role-based access control (RBAC)

Priority: Critical

Acceptance Criteria:

- Roles: Admin, Supervisor, Support, Dealer
- Permissions assigned per role
- Custom permission combinations
- Role change history logged
- Audit trail for permission changes

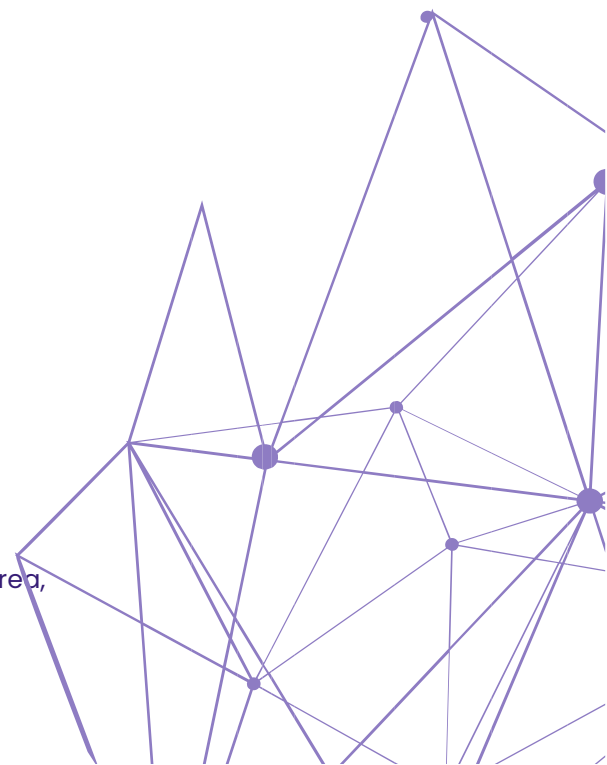
Requirement ID: FR-ADMIN-USER-003

Description: User KYC verification & approval

Priority: Critical

Acceptance Criteria:

- Admin reviews KYC documents
- Approve/reject with reason



- User notified of decision
- Rejected users can resubmit
- KYC status dashboard

Requirement ID: FR-ADMIN-USER-004

Description: Account suspension & reactivation

Priority: High

Acceptance Criteria:

- Suspend account with reason
- User cannot login when suspended
- Automatic reactivation after set period
- Manual reactivation option
- Suspension history recorded

Requirement ID: FR-ADMIN-USER-005

Description: User behavior tracking & analytics

Priority: Medium

Acceptance Criteria:

- Login history with timestamps
- Activity log per user
- Rental frequency tracked
- Spending patterns analyzed
- Fraud risk score calculated

Requirement ID: FR-ADMIN-USER-006

Description: Bulk user operations & reporting

Priority: Medium

Acceptance Criteria:

- Bulk email users
- Bulk SMS notifications
- User export with filters
- Segment users by criteria
- Activate/deactivate users in bulk

4.2.5 Financial Management

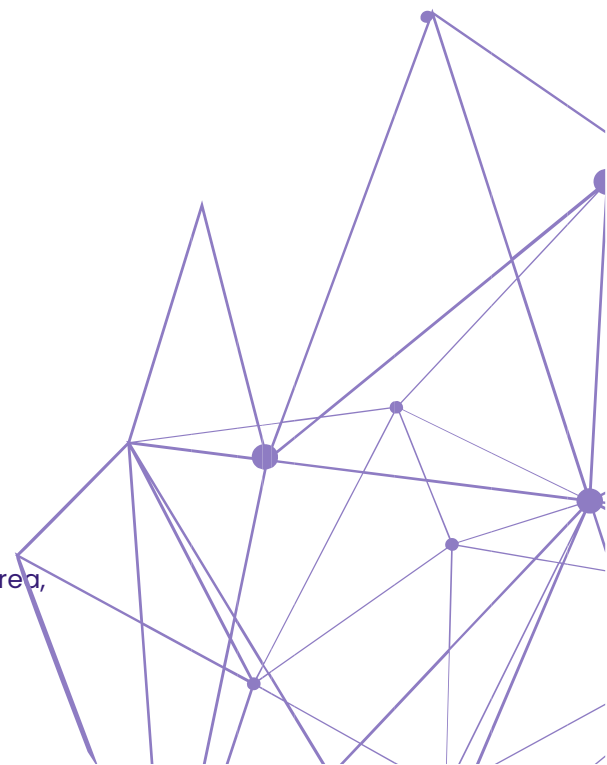
Requirement ID: FR-ADMIN-FIN-001

Description: Revenue tracking & reporting

Priority: Critical

Acceptance Criteria:

- Daily/weekly/monthly revenue shown



- Revenue by source (rental vs purchase)
- Revenue breakdown by station
- Revenue forecast for next month
- Revenue comparison with previous periods

Requirement ID: FR-ADMIN-FIN-002

Description: Commission management for dealers

Priority: High

Acceptance Criteria:

- Commission rates configurable per dealer
- Automatic calculation of commissions
- Monthly commission statement generated
- Payment due date tracked
- Dispute resolution process

Requirement ID: FR-ADMIN-FIN-003

Description: Payment settlement processing

Priority: High

Acceptance Criteria:

- Settlement schedule configured
- Automatic fund transfer initiated
- Settlement status tracked
- Failed settlement alert
- Reconciliation process

Requirement ID: FR-ADMIN-FIN-004

Description: Invoice generation & management

Priority: High

Acceptance Criteria:

- Auto-generate invoices for transactions
- Invoice number sequential
- GST calculation and display
- Invoice stored in database
- Invoice printable and downloadable

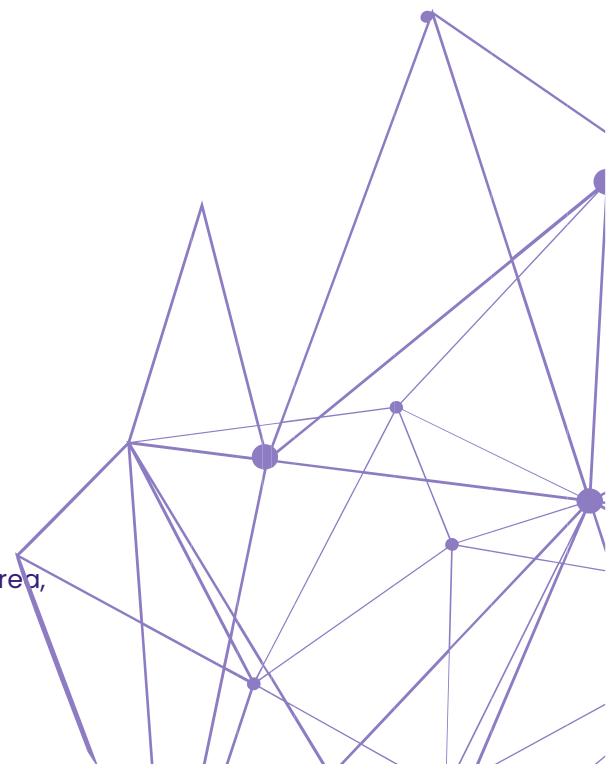
Requirement ID: FR-ADMIN-FIN-005

Description: Financial audit trails

Priority: High

Acceptance Criteria:

- All financial transactions logged



- User responsible for transaction recorded
- Transaction timestamp captured
- Amount before/after taxes shown
- Audit report exportable

Requirement ID: FR-ADMIN-FIN-006

Description: Profit margin analysis

Priority: Medium

Acceptance Criteria:

- Margin per station calculated
- Margin trend analysis (30/90 days)
- Margin by battery type
- Cost vs revenue comparison
- Profitability forecast

4.2.6 Support Ticketing System

Requirement ID: FR-ADMIN-TICKET-001

Description: Customer issue tracking

Priority: High

Acceptance Criteria:

- Create support ticket from user report
- Ticket status: Open, In Progress, Resolved, Closed
- Ticket priority: Low, Medium, High, Critical
- Ticket assigned to support agent
- Ticket updated with resolution

Requirement ID: FR-ADMIN-TICKET-002

Description: Automated ticket assignment

Priority: High

Acceptance Criteria:

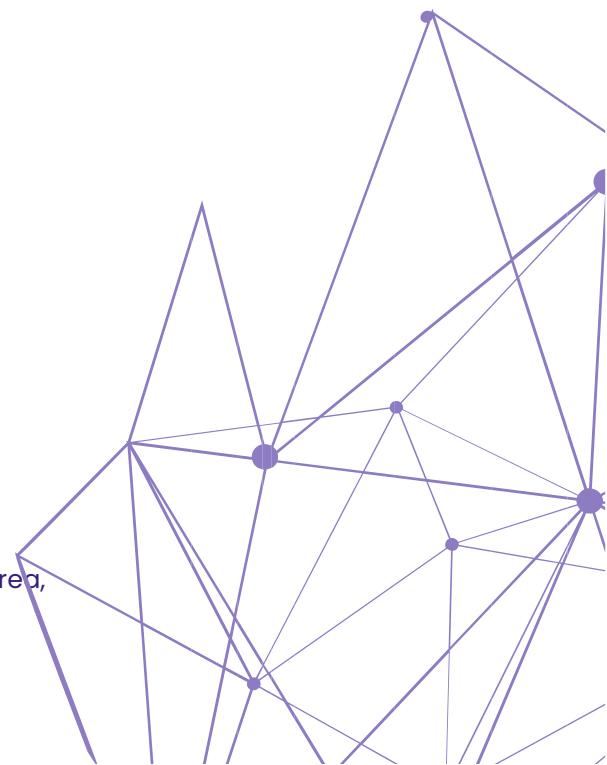
- Tickets assigned based on availability
- Load balancing across support team
- Reassignment option available
- Automatic assignment rules configurable
- Manual override allowed

Requirement ID: FR-ADMIN-TICKET-003

Description: Priority-based queue management

Priority: High

Acceptance Criteria:



- Queue sorted by priority
- SLA time shown (24hr for normal, 2hr for critical)
- Overdue tickets highlighted
- Queue statistics displayed
- Ticket aging report

Requirement ID: FR-ADMIN-TICKET-004

Description: Support team performance metrics

Priority: Medium

Acceptance Criteria:

- Average resolution time per agent
- Customer satisfaction rating per agent
- Tickets resolved per agent
- First contact resolution rate
- Quality score per agent

Requirement ID: FR-ADMIN-TICKET-005

Description: Knowledge base for common issues

Priority: Medium

Acceptance Criteria:

- Knowledge base searchable
- Article creation and editing
- Article categorization
- Self-service for customers
- Analytics on article views

4.3 Dealer Registration & Management Portal (Web) - Phase 1

4.3.1 Core Dealer Features

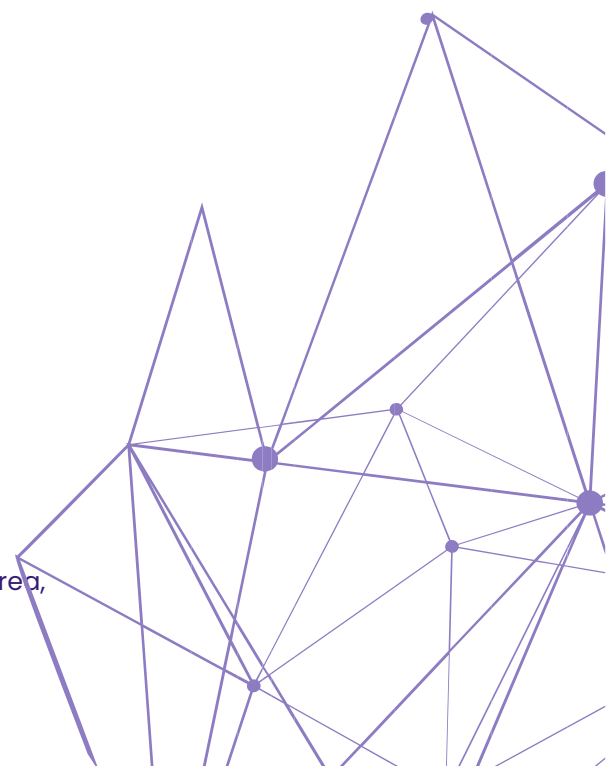
Requirement ID: FR-DEALER-001

Description: Dealer registration & verification

Priority: Critical

Acceptance Criteria:

- 8-stage onboarding process (7-8 days total)
- Digital application submission (Day 0-1: 15 mins)
- Automated checks (Day 1: 4 hours)
- KYC capture (Day 1-2: 30 mins)



- Manual review (Day 2-3: 1-2 hours)
- Field visit verification (Day 3-5: 1-2 hours)
- Final approval (Day 5-6: 30 mins)
- Setup & training (Day 6-7: 1-2 hours)
- Inventory handover (Day 7-8: 1-2 hours)

Requirement ID: FR-DEALER-002**Description:** Station setup & configuration**Priority:** High**Acceptance Criteria:**

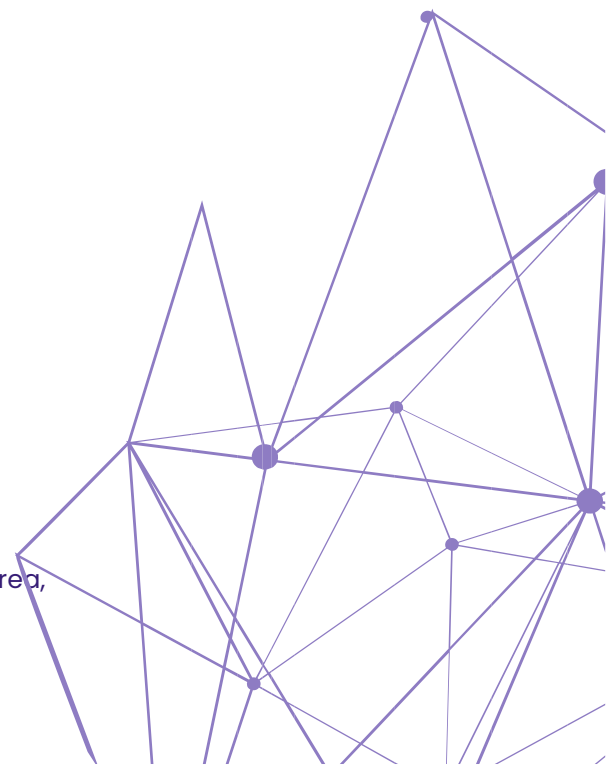
- Configure station name and location
- Add chargers and equipment details
- Set operating hours
- Configure capacity limits
- Upload station photos/documentation

Requirement ID: FR-DEALER-003**Description:** Inventory management**Priority:** High**Acceptance Criteria:**

- Add batteries to inventory
- Remove batteries from circulation
- Track battery health per station
- View stock levels real-time
- Forecast demand

Requirement ID: FR-DEALER-004**Description:** Sales tracking & revenue monitoring**Priority:** High**Acceptance Criteria:**

- Daily sales summary displayed
- Weekly sales report generated
- Monthly commission calculated
- Revenue trend chart
- Payment due dates visible

Requirement ID: FR-DEALER-005**Description:** Dealer-specific analytics dashboard**Priority:** High**Acceptance Criteria:**

- Dashboard customizable by dealer
- Sales metrics displayed
- Inventory metrics shown
- Customer satisfaction ratings
- Performance vs targets

Requirement ID: FR-DEALER-006

Description: Customer support for stations

Priority: High

Acceptance Criteria:

- Support ticket system
- Customer inquiries tracked
- Resolution timeline monitored
- Customer feedback captured
- Support performance metrics

Requirement ID: FR-DEALER-007

Description: Promotional campaign management

Priority: Medium

Acceptance Criteria:

- Create discount campaigns
- Schedule campaign dates
- Set discount percentage/amount
- Track campaign performance
- Automatic coupon generation

Requirement ID: FR-DEALER-008

Description: Document upload & management

Priority: High

Acceptance Criteria:

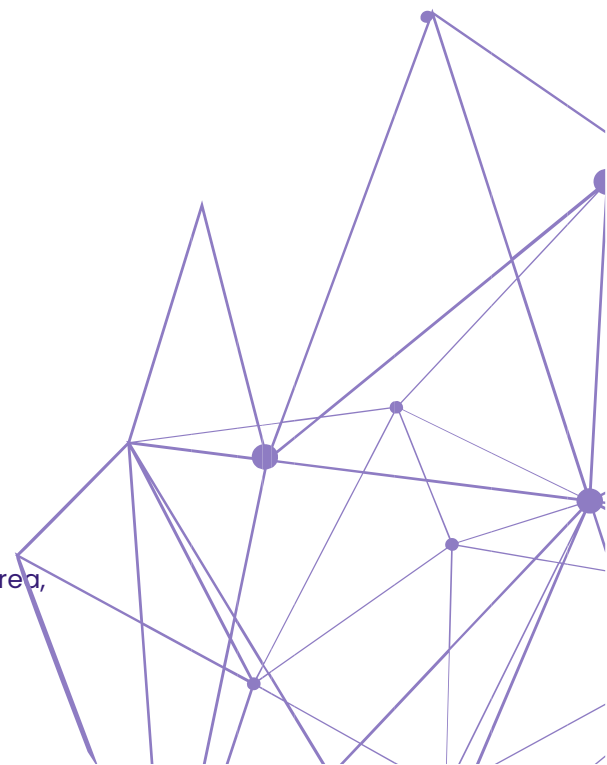
- Upload business licenses
- Upload GST certificates
- Upload insurance documents
- Store operational manuals
- Version control for documents

Requirement ID: FR-DEALER-009

Description: Real-time notification system

Priority: High

Acceptance Criteria:



- Alerts for low inventory
 - Alerts for maintenance due
 - Alerts for new bookings
 - Alerts for support tickets
 - Email and SMS notifications
-

4.4 Logistics & Delivery Platform (Web) - Phase 1

4.4.1 Core Logistics Features

Requirement ID: FR-LOGISTICS-001

Description: Delivery order management

Priority: Critical

Acceptance Criteria:

- Create delivery orders from admin
- Assign orders to delivery partners
- Track order status in real-time
- Order status: Pending, In Transit, Delivered, Failed
- Proof of delivery captured

Requirement ID: FR-LOGISTICS-002

Description: Real-time GPS tracking

Priority: Critical

Acceptance Criteria:

- Live location of delivery vehicle shown
- Battery marker shows location accuracy
- Delivery partner contact accessible
- ETA calculated automatically
- Delay alerts triggered

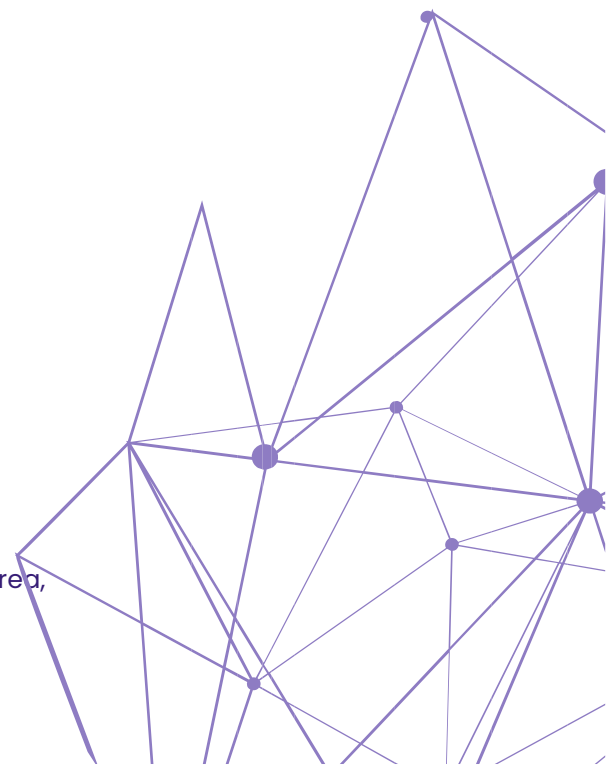
Requirement ID: FR-LOGISTICS-003

Description: Route optimization

Priority: High

Acceptance Criteria:

- Automatic route calculation (shortest/fastest)
- Multi-stop route planning
- Traffic condition consideration



- Real-time route recalculation
- Route history and analytics

Requirement ID: FR-LOGISTICS-004**Description:** Proof of delivery (PoD)**Priority:** High**Acceptance Criteria:**

- Photo capture at delivery location
- Signature/OTP verification
- Timestamp recorded
- Recipient details captured
- PoD stored in system

Requirement ID: FR-LOGISTICS-005**Description:** Driver assignment & management**Priority:** High**Acceptance Criteria:**

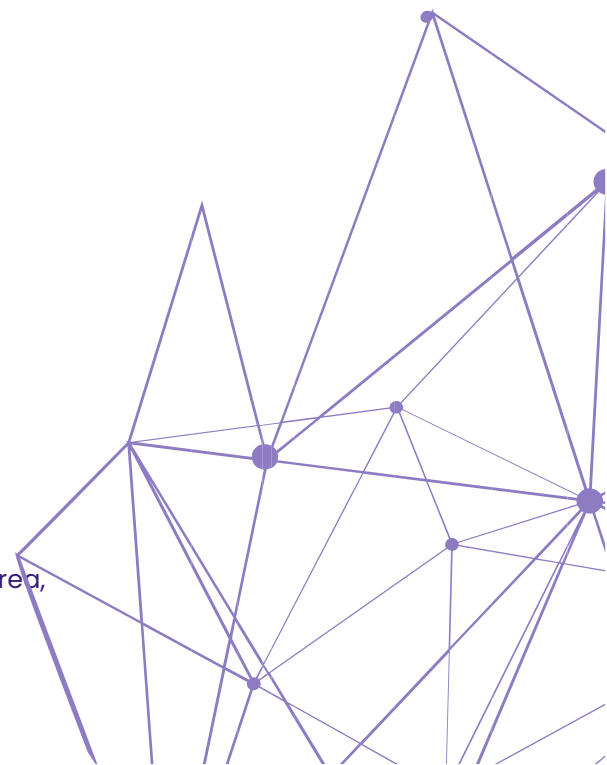
- Create driver profiles
- Track driver performance
- Rate driver reliability
- Monitor delivery metrics
- Driver availability status

Requirement ID: FR-LOGISTICS-006**Description:** Delivery performance metrics**Priority:** High**Acceptance Criteria:**

- On-time delivery percentage
- Average delivery time
- Failed delivery count
- Customer satisfaction rating
- Efficiency score

Requirement ID: FR-LOGISTICS-007**Description:** Customer communication**Priority:** High**Acceptance Criteria:**

- SMS with delivery updates
- Delivery tracking link shared
- Delivery partner phone shared



- Flexible delivery scheduling
- Customer confirmation before delivery

Requirement ID: FR-LOGISTICS-008

Description: Return & reverse logistics

Priority: High

Acceptance Criteria:

- Return order initiation
- Return pickup scheduling
- Return tracking
- Return delivery confirmation
- Return processing for refunds

Requirement ID: FR-LOGISTICS-009

Description: Inventory sync across stations

Priority: High

Acceptance Criteria:

- Inventory transferred between stations
- Stock updates automatic
- Transfer history tracked
- Discrepancy alerts
- Reconciliation process

4.5 Phase 2 Features - Advanced Analytics & AI/ML (Months 4-6)

4.5.1 Fraud Detection Engine

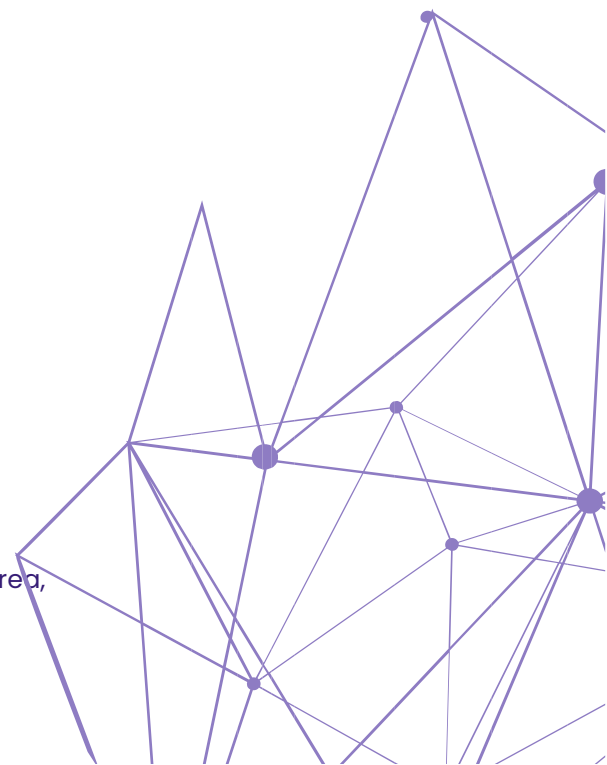
Requirement ID: FR-FRAUD-001

Description: Multi-layer fraud detection system

Priority: Critical

Acceptance Criteria:

- 95%+ fraud detection accuracy
- Real-time fraud scoring (0-100)
- Multi-layer verification (PAN, Aadhaar, GST, Phone)
- Auto-approve: Risk score 0-20
- Hold for review: Risk score 20-60
- Auto-reject: Risk score 60+
- Manual override with documentation



Requirement ID: FR-FRAUD-002**Description:** KYC verification scoring**Priority:** Critical**Acceptance Criteria:**

- PAN name match (cross-verify vs Aadhaar, GST, Application)
- GST legal status verification (real-time API)
- Phone fraud score from third-party database
- Duplicate detection (PAN/Aadhaar/Phone/Email)
- Background screening (police/court records)

Requirement ID: FR-FRAUD-003**Description:** Behavioral anomaly detection**Priority:** High**Acceptance Criteria:**

- ML model detects unusual rental patterns
- Large unusually rental requests flagged
- Multiple failed payments tracked
- Rapid account creation attempts blocked
- Geographic anomalies detected

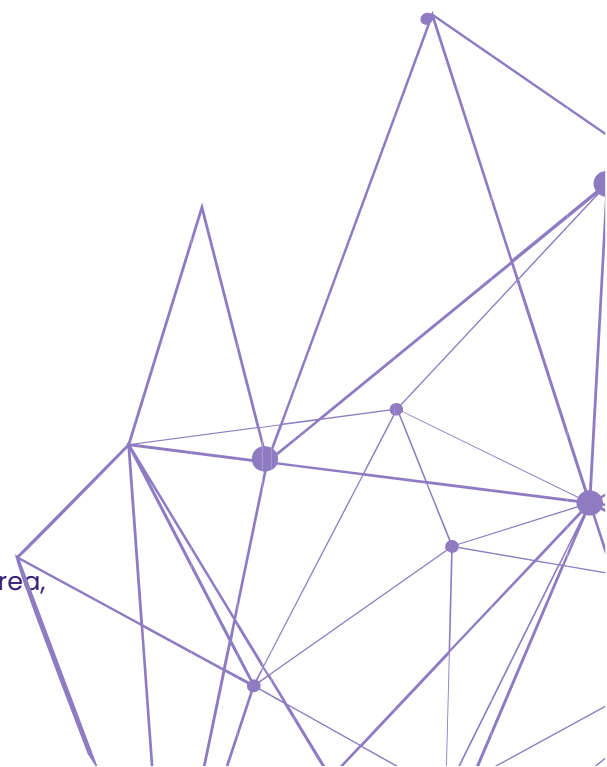
Requirement ID: FR-FRAUD-004**Description:** Device fingerprinting**Priority:** Medium**Acceptance Criteria:**

- Unique device ID generated
- IP address tracked
- Device model and OS captured
- Multiple accounts per device detected
- Compromised device alerts

4.5.2 AI/ML Predictive Models

Requirement ID: FR-AI-001**Description:** Demand forecasting system**Priority:** High**Acceptance Criteria:**

- 30-day demand forecast per station
- Forecast accuracy >85%
- Seasonal patterns considered
- Weather impact analyzed



- Events impact considered

Requirement ID: FR-AI-002

Description: Revenue optimization engine

Priority: High

Acceptance Criteria:

- Dynamic pricing recommendations
- Peak vs off-peak pricing
- Demand-based price adjustments
- Revenue increase projections
- A/B testing framework

Requirement ID: FR-AI-003

Description: Churn prediction model

Priority: Medium

Acceptance Criteria:

- Predict customer churn probability
- High-risk customers identified
- Retention recommendations
- Intervention campaign setup
- Model accuracy >80%

Requirement ID: FR-AI-004

Description: Customer segmentation

Priority: Medium

Acceptance Criteria:

- Segment customers by behavior
- RFM analysis (Recency, Frequency, Monetary)
- Personalized offers per segment
- Segment performance tracking
- Dynamic segment creation

4.5.3 Advanced Analytics Dashboard

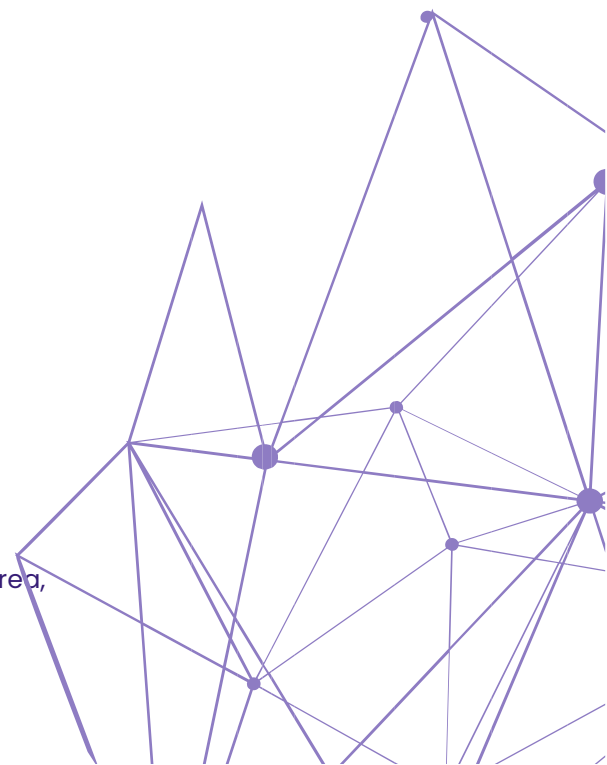
Requirement ID: FR-ANALYTICS-001

Description: Neo4j graph database implementation

Priority: High

Acceptance Criteria:

- Complex relationship queries
- Customer-Battery-Station relationships
- Dealer-Station-Inventory relationships



- Transaction-User relationship analysis
- Network analysis and visualization

Requirement ID: FR-ANALYTICS-002**Description:** TimescaleDB time-series analytics**Priority:** High**Acceptance Criteria:**

- Store time-series data for all events
- Query optimization for time ranges
- Compression of old data
- Retention policy (3 years default)
- Time-series charts and analysis

Requirement ID: FR-ANALYTICS-003**Description:** Business intelligence reports**Priority:** High**Acceptance Criteria:**

- Executive dashboard with key metrics
- Customizable report builder
- Scheduled report emails
- Data export to multiple formats
- Drill-down capability

Requirement ID: FR-ANALYTICS-004**Description:** Real-time data pipeline**Priority:** High**Acceptance Criteria:**

- Event streaming from IoT devices
- Real-time aggregations
- Sub-second latency for dashboards
- Fault-tolerant data processing
- Auto-scaling based on load

5 Non-Functional Requirements

5.1 Performance Requirements

Requirement ID	Metric	Target	Priority
NFR-PERF-001	API response time	<200ms (95th percentile)	Critical

NFR-PERF-002	App load time	<3 seconds	High
NFR-PERF-003	Dashboard load time	<2 seconds	High
NFR-PERF-004	Database query time	<500ms (95th percentile)	High
NFR-PERF-005	Concurrent user capacity	10,000+ users	Critical
NFR-PERF-006	Transaction throughput	1,000 TPS	High
NFR-PERF-007	Uptime SLA	99.9%	Critical
NFR-PERF-008	Mean time to recovery (MTTR)	<15 minutes	High
NFR-PERF-009	Batch processing time	<5 minutes (for daily jobs)	Medium
NFR-PERF-010	Mobile app offline capability	30 minutes	Medium

5.2 Security Requirements

Requirement ID: NFR-SEC-001

Description: Data encryption at rest and in transit

Priority: Critical

Acceptance Criteria:

- AES-256 encryption for sensitive data
- TLS 1.2+ for all data in transit
- HTTPS enforced everywhere
- Certificate pinning for mobile app
- Regular encryption key rotation

Requirement ID: NFR-SEC-002

Description: Authentication & authorization

Priority: Critical

Acceptance Criteria:

- OAuth2/JWT implementation
- MFA optional for users
- Biometric login support
- Session timeout after 30 minutes
- Logout on all devices option

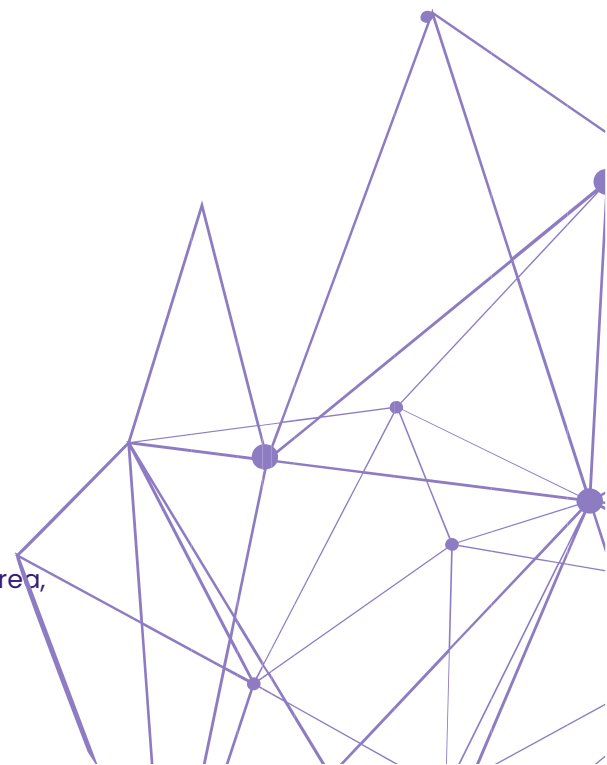
Requirement ID: NFR-SEC-003

Description: PCI-DSS compliance

Priority: Critical

Acceptance Criteria:

- Level 1 compliance achieved
- No card data stored on servers



- Tokenization implemented
- 3D Secure for card transactions
- Annual PCI-DSS audit passed

Requirement ID: NFR-SEC-004

Description: Data privacy & GDPR compliance

Priority: High

Acceptance Criteria:

- User consent for data collection
- Right to data deletion honored
- Data export functionality provided
- Privacy policy accessible
- Annual privacy audit

Requirement ID: NFR-SEC-005

Description: Regular security updates

Priority: High

Acceptance Criteria:

- Monthly security patches
- Vulnerability scanning automated
- Penetration testing quarterly
- Security incident response plan
- Staff security training annual

Requirement ID: NFR-SEC-006

Description: DDoS protection

Priority: High

Acceptance Criteria:

- DDoS mitigation service deployed
- Rate limiting implemented
- IP blacklist/whitelist
- Automatic traffic filtering
- Attack alerts and reporting

5.3 Scalability Requirements

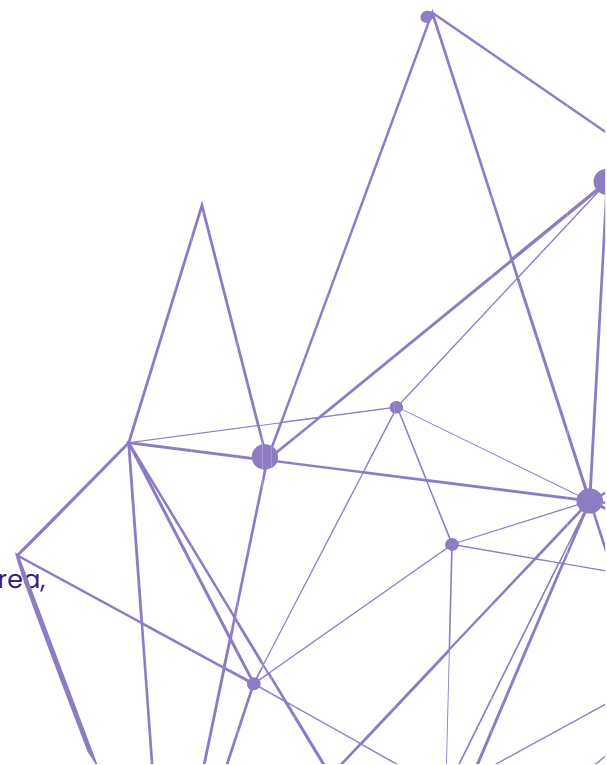
Requirement ID: NFR-SCALE-001

Description: Horizontal scaling

Priority: High

Acceptance Criteria:

- Load balancer distributes traffic



- Auto-scaling based on CPU/memory
- Stateless application design
- Session replication across servers
- Database read replicas

Requirement ID: NFR-SCALE-002**Description:** Database scalability**Priority:** High**Acceptance Criteria:**

- Sharding strategy implemented
- Replication configured
- Backup and recovery procedures
- Query optimization ongoing
- Capacity planning quarterly

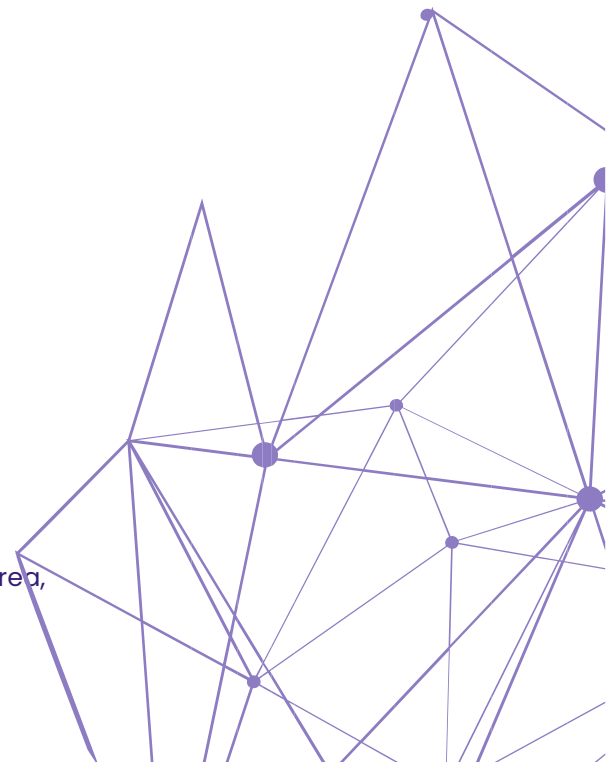
Requirement ID: NFR-SCALE-003**Description:** Storage capacity**Priority:** High**Acceptance Criteria:**

- Object storage (S3) for files
- Automatic archival of old data
- Data retention policy enforced
- Capacity planning for 3 years
- Cost optimization for storage

5.4 Availability & Reliability

Requirement ID: NFR-AVAIL-001**Description:** System uptime (99.9% SLA)**Priority:** Critical**Acceptance Criteria:**

- 99.9% uptime over 30 days
- Planned maintenance <4 hours/month
- Automatic failover implemented
- Multiple data center deployment
- Regular disaster recovery drills

Requirement ID: NFR-AVAIL-002**Description:** Disaster recovery plan**Priority:** Critical**Acceptance Criteria:**

- RPO (Recovery Point Objective): <1 hour
- RTO (Recovery Time Objective): <15 minutes
- Backup frequency: Every 6 hours
- Backup location: Geo-redundant
- DR drill quarterly

Requirement ID: NFR-AVAIL-003

Description: Monitoring & alerting

Priority: High

Acceptance Criteria:

- Real-time monitoring of all systems
- Alert sent within 2 minutes of issue
- 24/7 incident response team
- Alert escalation procedures
- Incident tracking and reporting

5.5 Usability & Accessibility

Requirement ID: NFR-USAB-001

Description: Mobile app responsiveness

Priority: High

Acceptance Criteria:

- Works on iOS 12+ and Android 8+
- Responsive design for all screen sizes
- Touch-friendly interface (min 44x44px)
- Fast app startup (<3 seconds)
- Minimal battery drain

Requirement ID: NFR-USAB-002

Description: Web accessibility (WCAG 2.1 AA)

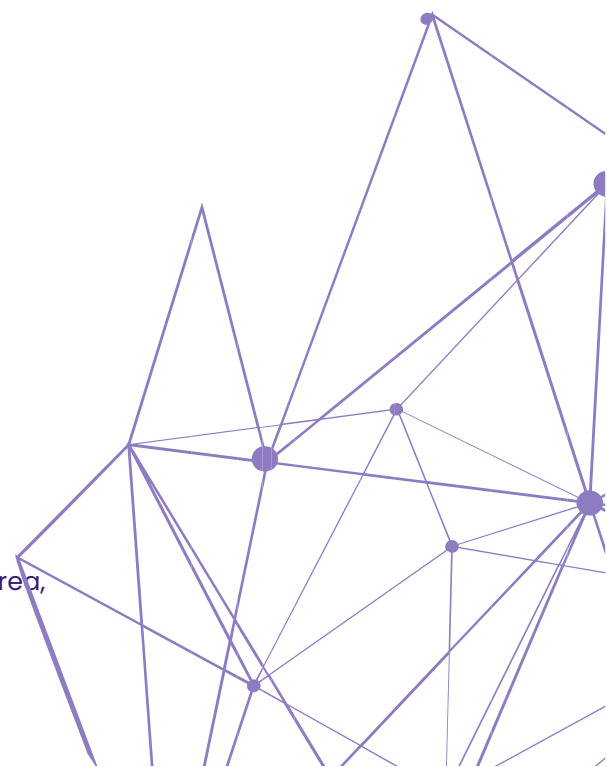
Priority: High

Acceptance Criteria:

- Color contrast ratio 4.5:1 for text
- Keyboard navigation support
- Screen reader compatible
- Alt text for all images
- Clear focus indicators

Requirement ID: NFR-USAB-003

Description: Multi-language support



Priority: Medium**Acceptance Criteria:**

- English and Hindi support
- Language selection option
- Right-to-left (RTL) for future languages
- Date/time localization
- Currency localization

Requirement ID: NFR-USAB-004**Description:** Offline capability**Priority:** Medium**Acceptance Criteria:**

- View past rentals offline
- View station list offline
- Sync when online resumed
- Clear offline indicator
- 30-minute offline window

6 Technical Architecture

6.1 System Architecture Overview

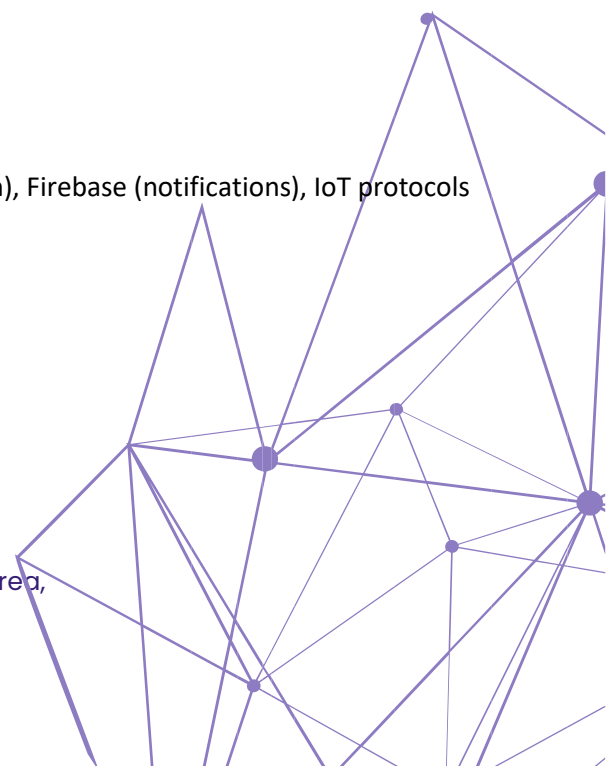
Architecture Style: Microservices architecture with event-driven components**Core Components:**

- **Frontend Layer:** React Native (mobile), React.js (web)
- **API Layer:** Node.js/Express REST APIs
- **Database Layer:** MongoDB (primary), PostgreSQL (relational), Redis (cache)
- **Real-time Layer:** [Socket.io](#) for live updates
- **Analytics Layer:** Neo4j, TimescaleDB, data pipeline
- **Infrastructure:** AWS cloud services with Docker/Kubernetes
- **Third-party Services:** Razorpay (payments), Google Maps (location), Firebase (notifications), IoT protocols (MQTT)

6.2 Technology Stack

Frontend:

- React Native 0.70+ for iOS & Android
- React 18+ for web dashboards
- Redux/Context API for state management



- Tailwind CSS for styling
- Google Maps API integration

Backend:

- Node.js 18+ with Express.js 4.x
- REST API architecture
- JWT & OAuth2 for authentication
- [Socket.io](#) for real-time communication

Database:

- MongoDB 5.0+ (primary NoSQL)
- PostgreSQL 14+ (relational data)
- Redis 7.0+ (caching/sessions)
- Neo4j 5.0+ (graph data - Phase 2)
- TimescaleDB 2.0+ (time-series - Phase 2)

Infrastructure:

- AWS EC2 for compute
- AWS RDS for managed databases
- AWS S3 for file storage
- AWS CloudFront for CDN
- Docker containerization
- Kubernetes orchestration (Phase 2)

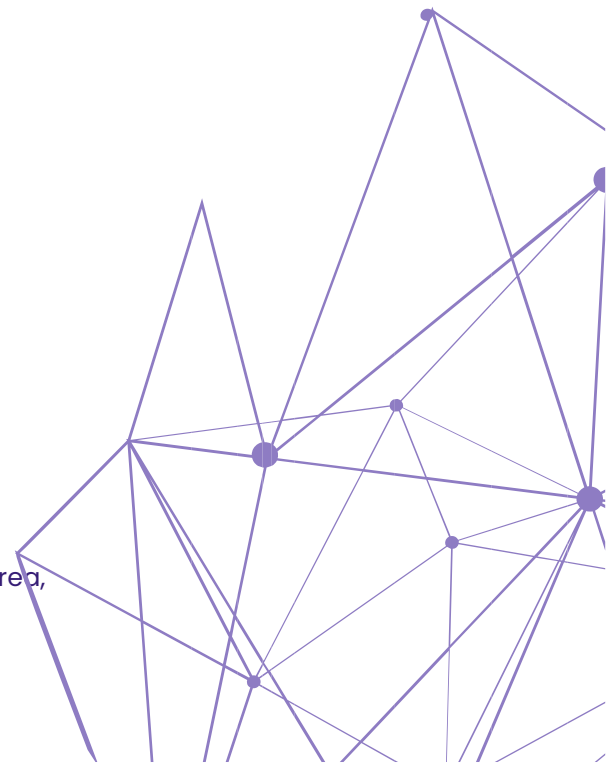
Integrations:

- Razorpay payment gateway
- Google Maps API
- Firebase Cloud Messaging
- Twilio SMS
- IoT MQTT protocol

6.3 API Design & Standards

REST API Principles:

- RESTful endpoint design
- Versioning: `/api/v1/`, `/api/v2/`
- Standard HTTP status codes
- JSON request/response format



- Rate limiting: 1000 requests/minute

API Documentation:

- Swagger/OpenAPI specification
- Interactive API documentation
- Request/response examples
- Error code documentation
- Authentication guide

6.4 Data Storage & Management

Data Models:

- User model (profile, KYC, wallet)
- Battery model (specs, health, location)
- Station model (info, inventory, hours)
- Rental model (history, tracking)
- Transaction model (payments, invoices)
- Dealer model (profile, commission)

Data Retention Policy:

- Active data: 3 years
- Archived data: 7 years (compliance)
- Logs: 1 year
- Transactions: Permanent
- User data: Until account deletion + 30 days

7 Integration Requirements

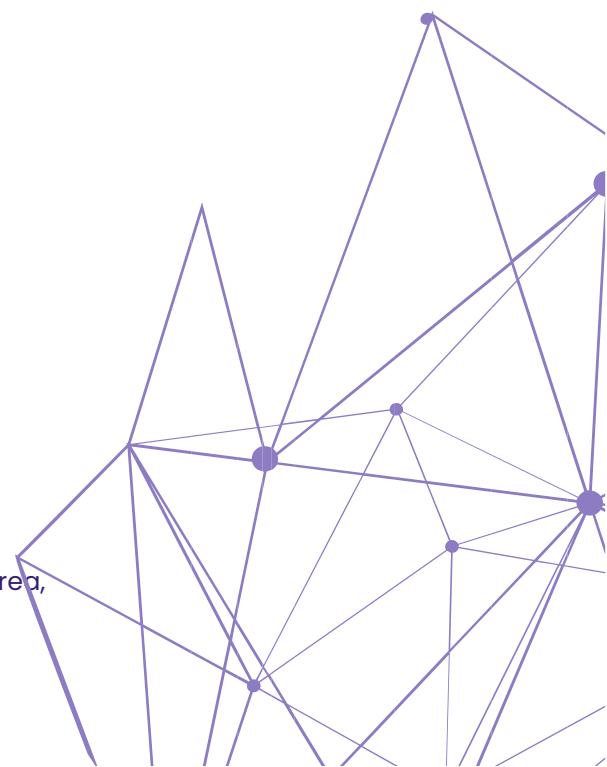
7.1 Third-Party Integrations

Payment Gateway - Razorpay:

- Merchant account setup
- Test and production keys
- PCI-DSS Level 1 compliance
- Webhook for transaction updates
- Refund API integration

Location Services - Google Maps:

- Maps API for map display



- Geolocation API
- Directions API for routing
- Places API for station search
- API quota: 50,000 daily

Notifications - Firebase:

- FCM setup for push notifications
- Topic-based messaging
- Rich media notifications
- Analytics tracking
- Quota: 10,000,000 messages/day

SMS - Twilio:

- SMS API for OTP and alerts
- Bulk SMS capability
- Delivery reports
- Quota: 100,000 SMS/month

KYC Verification:

- Aadhaar e-KYC API
- PAN verification (NSDL)
- GST API verification
- Video KYC service provider

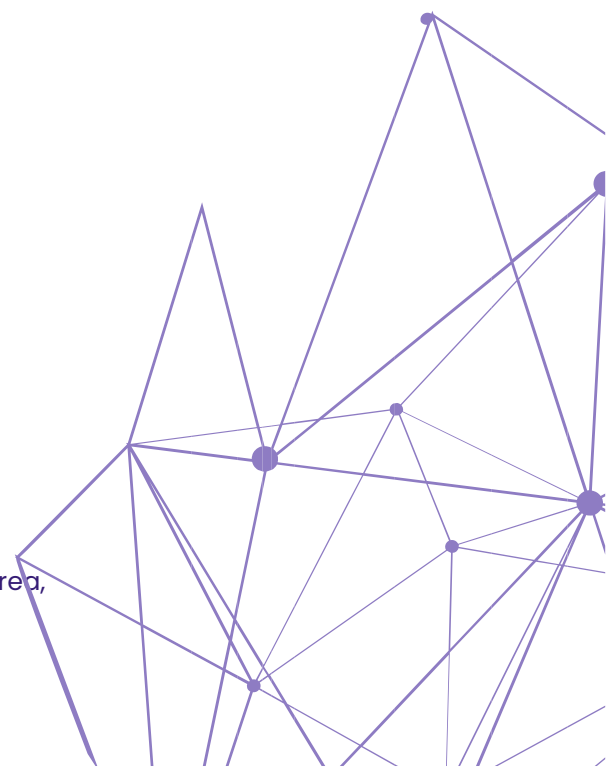
IoT Device Communication:

- MQTT protocol for device communication
- Battery data ingestion
- Real-time telemetry
- Device command execution

7.2 IoT Integration

Requirement ID: FR-IOT-001**Description:** IoT device communication framework**Priority:** Critical**Acceptance Criteria:**

- MQTT connection to IoT devices
- Bidirectional communication
- Data transmission every 4 hours



- Remote lock capability
- Theft detection protocols
- Serial number mapping
- RFID tracking integration
- Geo-fencing alerts
- Battery health data collection
- Temperature monitoring

8 Project Phases & Timeline

8.1 Phase 1 Timeline (Months 1-3): Core Platform

Month	Week	Deliverables	Status
Month 1	1-4	Project setup, Database design, API development (50%), UI mockups	In Progress
Month 2	5-8	API development (100%), Mobile app (70%), Admin dashboard (60%), Testing	Planned
Month 3	9-12	App completion (100%), Beta testing, Dealer portal launch, Production deployment	Planned

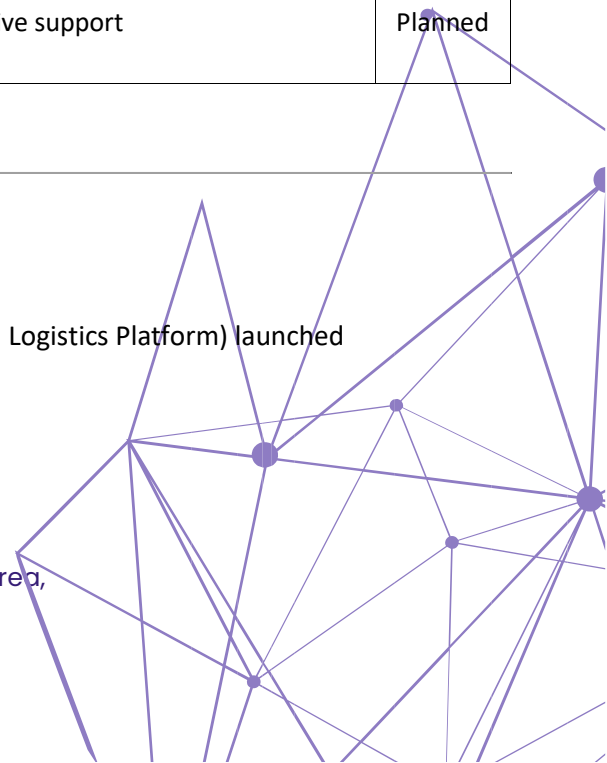
8.2 Phase 2 Timeline (Months 4-6): Advanced Features

Month	Week	Deliverables	Status
Month 4	13-16	AI/ML models, Analytics infrastructure, Performance optimization, Security hardening	Planned
Month 5	17-20	Advanced features, Fraud detection system, Business intelligence, Integration testing	Planned
Month 6	21-24	Final optimization, Load testing, Documentation, Go-live support	Planned

9 Success Criteria & Acceptance

9.1 Phase 1 Success Criteria

- ✓ All 4 applications (Mobile App, Admin Dashboard, Dealer Portal, Logistics Platform) launched
- ✓ 100+ test users completed transactions successfully
- ✓ 99.5% system uptime achieved



- ✓ All payment transactions processed securely
- ✓ GPS tracking accurate within 10 meters
- ✓ Customer support responding within 2 hours
- ✓ Zero critical security vulnerabilities
- ✓ Complete user documentation available

9.2 Phase 2 Success Criteria

- ✓ AI/ML models achieving >90% accuracy
- ✓ Fraud detection system preventing 95%+ fraudulent transactions
- ✓ Advanced analytics dashboard processing 1M+ data points
- ✓ System handling 10,000+ concurrent users
- ✓ 99.9% uptime SLA maintained
- ✓ All advanced features fully operational
- ✓ Platform ready for public launch

10 Risk Management

10.1 Identified Risks

Risk: Technology integration complexity

Probability: High | **Impact:** High

Mitigation: Early integration testing, vendor support agreements, dedicated integration team

Risk: Dealer KYC verification delays

Probability: Medium | **Impact:** High

Mitigation: Third-party KYC service provider, parallel processing, dedicated KYC team

Risk: IoT device connectivity issues

Probability: Medium | **Impact:** High

Mitigation: MQTT redundancy, fallback communication, device health monitoring

Risk: Payment gateway failures

Probability: Low | **Impact:** Critical

Mitigation: Dual payment gateway integration, retry mechanism, manual fallback

Risk: Compliance and regulatory changes

Probability: Medium | **Impact:** High

Mitigation: Legal team consultation, compliance monitoring, documentation

11 Post-Launch Support

11.1 Support Plan

24/7 Technical Support

- Critical issues: Response within 15 minutes
- High priority issues: Response within 2 hours
- Medium priority: Response within 24 hours
- Low priority: Response within 48 hours

Maintenance Schedule

- Monthly security updates
- Quarterly feature releases
- Weekly database optimization
- Bi-weekly performance reviews

Monitoring & Analytics

- Real-time system monitoring
- Daily health reports
- Weekly performance reports
- Monthly business analytics

User Support

- Email support: support@wezu.com
- Chat support: 24/7 AI + human agents
- Phone support: Business hours
- Knowledge base with video tutorials

12 Glossary & Definitions

Term	Definition
KYC	Know Your Customer - identity verification process
GST	Goods and Services Tax - Indian tax system
PAN	Permanent Account Number - tax identification
Aadhaar	Unique 12-digit identity number in India
IoT	Internet of Things - connected devices
MQTT	Message Queuing Telemetry Transport - IoT protocol

API	Application Programming Interface
JWT	JSON Web Token - authentication mechanism
RPO	Recovery Point Objective - data backup frequency
RTO	Recovery Time Objective - system recovery time
SLA	Service Level Agreement - uptime guarantee
WCAG	Web Content Accessibility Guidelines

13 Approval & Sign-Off

Prepared by: Project Management Team

Version: 1.0

Date: December 17, 2025

Status: Ready for Review

Stakeholder Approvals:

Stakeholder	Role	Signature	Date
[Name]	Project Sponsor	_____	_____
[Name]	Technical Lead	_____	_____
[Name]	Product Manager	_____	_____
[Name]	QA Lead	_____	_____

13.1 CLOUD APPLICATION SOLUTION DEVELOPMENT

13.2 CLAUSE LIBRARY & MSA FRAMEWORK

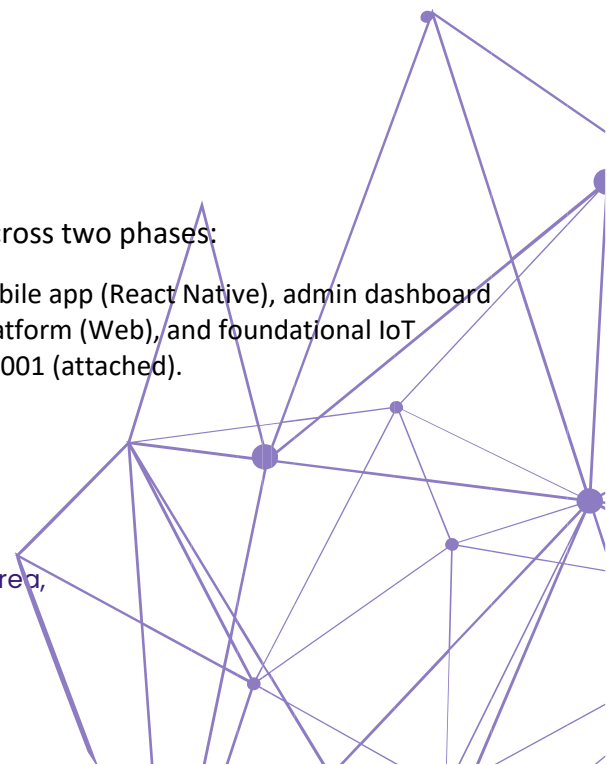
13.3 SCOPE & DELIVERABLES CLAUSES

13.3.1 Scope of Work & Project Structure

13.3.1.1 *CLAUSE: Multi-Phase Project Scope*

The Vendor shall deliver the WEZU Energy Monetization Platform across two phases:

- **Phase 1 (Months 1–3):** Core platform encompassing customer mobile app (React Native), admin dashboard (React Web), dealer registration portal (Web), logistics delivery platform (Web), and foundational IoT integration. Deliverables outlined in Section 3.1–3.4 of WEZU-SRS-001 (attached).



- **Phase 2 (Months 4–6):** Advanced analytics, AIML-driven fraud detection engine, predictive demand forecasting, churn prediction, and revenue optimization models. Deliverables outlined in Section 3.5 of WEZU-SRS-001.

All Phase 1 deliverables are mandatory for production go-live. Phase 2 features are subject to a separate Statement of Work (SOW) and optional engagement unless explicitly incorporated into this Agreement via written amendment.

13.3.2 Detailed Deliverables & Acceptance Criteria

13.3.2.1 *CLAUSE: Functional Requirement Mapping*

For each Functional Requirement (FR) and Non-Functional Requirement (NFR) in WEZU-SRS-001, the Vendor shall:

1. Implement the feature/capability as specified in the Acceptance Criteria column.
2. Provide evidence of acceptance testing and proof of conformity (test execution logs, screenshots, defect reports if any).
3. Deliver corresponding technical documentation (API specs, data schemas, configuration guides).

Traceability matrix linking each FR/NFR ID to SOW deliverables and acceptance test cases shall be maintained and updated weekly and made available to the Customer in the project repository.

13.3.3 Environments & Infrastructure

13.3.3.1 *CLAUSE: Standard Environments*

Vendor shall provision and maintain the following environments:

Environment	Purpose	Data	Uptime SLA
Development (Dev)	Code development, feature integration	Anonymized/test data	Best effort
Quality Assurance (QA)	Functional & regression testing	Test data replicas	95% during test hours
Staging (Pre-Prod)	UAT, performance testing, go-live dress rehearsals	Production-like data (anonymized)	99%
Production (Prod)	Live customer-facing platform	Real customer data (encrypted)	99.9% (as per NFR-AVAIL-001)

All environments shall be isolated, with role-based access control, separate databases, and clear promotion workflows (Dev → QA → Staging → Prod).

13.3.4 Change Management

13.3.4.1 *CLAUSE: Formal Change Request Process*

Any modification to scope, requirements, technology stack, timelines, or deliverables must follow this process:

1. **Initiation:** Customer or Vendor submits Change Request (CR) form with title, description, business justification, and proposed solution.
2. **Impact Analysis:** Vendor completes impact analysis within 3 business days, documenting:
 - Effort (hours/days required)
 - Schedule impact (timeline shift)
 - Cost impact (additional fees, if any)
 - Risk assessment
 - Dependence on other work
3. **Review & Approval:** Steering Committee reviews CR and approves/rejects/defers within 5 business days.
4. **Implementation:** Upon approval, CR is added to project backlog with agreed date, and tracked in the weekly status report.

Out-of-Scope Clarification:

Minor bug fixes, configuration updates, and documentation revisions within the agreed feature set are in-scope. Requests involving new features, third-party integrations not in SRS, or technology changes require a CR.

13.4 COMMERCIAL & GOVERNANCE CLAUSES

13.4.1 Pricing & Payment Terms

13.4.1.1 *CLAUSE: Phase 1 Fixed Price Milestone Payments*

Total Phase 1 Fee: ₹[INSERT AMOUNT] (all-inclusive of taxes, travel, and overheads)

Milestone Payment Schedule:

- Milestone 1 (Design & Architecture Sign-Off): 20% on Day 30
- Milestone 2 (Alpha Build - Core Features UAT): 30% on Day 60
- Milestone 3 (Beta Build - Full Feature Set): 30% on Day 85
- Milestone 4 (Production Go-Live & 30-Day Warranty): 20% on Day 120 post go-live

Payment Terms: Net 15 days from invoice date

Invoice Trigger: Steering Committee sign-off on milestone deliverables

13.4.1.2 *CLAUSE: Phase 2 Optional Engagement*

Phase 2 (AIML/Analytics) shall be quoted separately upon completion of Phase 1. Vendor shall provide a separate SOW with: - Detailed feature list for each FR/NFR in Section 3.5 (WEZU-SRS-001) - Resource allocation and timeline - Estimated fee and payment schedule - Option expiry date (e.g., 30 days after Phase 1 go-live)

13.4.2 Out-of-Scope & Change Costs

13.4.2.1 CLAUSE: Change Request Fees

Approved CRs shall be charged on a Time & Materials basis: - Senior Developer: ₹[INSERT]/hour - Developer: ₹[INSERT]/hour - QA Engineer: ₹[INSERT]/hour - DevOps/Infra: ₹[INSERT]/hour

All T&M work requires a signed CR with estimated hours (e.g., “10–15 hours estimated effort”) and written approval before work commences. Actual hours shall be tracked weekly and reported in status updates.

13.4.3 Project Governance

13.4.3.1 CLAUSE: Steering Committee & Governance Cadence

The Steering Committee shall comprise: - **Customer:** Project Sponsor, Technical Lead, Business Analyst (3 members) - **Vendor:** Project Manager, Technical Architect, Senior Developer (3 members) **Service Provider:** Project Manager, Technical Architect, Senior Developer (3 members)

Governance Meetings: - **Weekly Project Call:** Every Monday, 10:00–10:45 AM IST. Agenda: status, blockers, risks, changes. Minutes recorded and circulated within 24 hours. - **Monthly Steering Review:** First Friday of each month. Agenda: milestone progress, budget, risks, executive updates. - **Ad Hoc Escalation:** Emergency calls scheduled within 4 business hours if critical blockers arise.

All attendees are expected to review agenda items before the call and come prepared to make decisions.

13.4.4 Escalation Matrix

13.4.4.1 CLAUSE: Issue Escalation Levels

Issue Type	L1 (Team)	L2 (Management)	L3 (Steering)	Escalation Timeline
Technical blocker	Project Manager/Technical Lead	Vendor CTO / Customer IT Head	Steering Committee	Within 1 day if unresolved at L1
Scope dispute	Project Manager	Commercial/Legal	Steering Committee	Within 2 days if unresolved at L1
Schedule risk	Project Manager	PMO/Account Manager	Steering Committee	Within 1 day of identification
Budget overrun	Project Manager	Finance/Account Manager	Steering Committee	Within 3 days
Security/Compliance incident	Security Officer	CTO/Chief Compliance	Steering Committee	Immediately (within 4 hours)

Unresolved issues at L3 shall be escalated to the respective executive sponsors for final decision.

13.5 PERFORMANCE & SLA CLAUSES

13.5.1 Availability & Uptime SLA

13.5.1.1 CLAUSE: Production Uptime Guarantee

Vendor warrants that the WEZU platform (production environment) shall maintain a minimum monthly uptime of **99.9%**, measured as:

$$\text{Uptime \%} = (\text{Total Minutes in Month} - \text{Downtime Minutes}) / \text{Total Minutes in Month} \times 100$$

Measurement Period: Calendar month (1st to end of month, IST time zone)

Exclusions from Downtime: - Scheduled maintenance windows (maximum 4 hours per month, with 48-hour notice) - Force majeure events (earthquakes, government action, telecommunications outages) - Customer-caused downtime or data corruption - Third-party service failures (payment gateway, SMS provider, cloud infrastructure provider outages >30 minutes)

Proof of Uptime: Vendor shall provide a monthly uptime report within 5 business days of month-end, certified by a third-party monitoring service (e.g., Uptime Robot, Data Dog) and signed by Vendor's authorized representative.

13.5.2 Performance KPIs

13.5.2.1 CLAUSE: Target Performance Metrics

The platform meets the following performance benchmarks under normal production load (defined as 10,000 concurrent users):

KPI	Target	Measurement	Frequency
API Response Time (95th percentile)	≤ 200 ms	Backend API calls excluding network latency	Daily
Mobile App Load Time (iOS/Android)	≤ 3 seconds	Time to first render content on 4G network	Daily
Admin Dashboard Load Time	≤ 2 seconds	Page load time on Chrome/Firefox from IST	Daily
Database Query Time (95th percentile)	≤ 500 ms	SELECT queries on production database	Daily
Concurrent User Capacity	≥ 10,000 users	Simultaneous active sessions without degradation	Weekly

KPI	Target	Measurement	Frequency
Transaction Throughput	≥ 1,000 TPS	Transactions per second during peak usage	Weekly
Batch Job Completion	≤ 5 minutes	Daily inventory sync, revenue rollup, fraud scoring	Daily
Mobile App Offline Capability	≥ 30 minutes	Functionality available without internet (cached data)	Monthly

Performance Testing: Vendor shall conduct load tests monthly, with results shared with the Customer. If targets are not met, Vendor shall provide a root-cause analysis and remediation plan within 5 business days.

13.5.3 Incident Response SLAs

13.5.3.1 *CLAUSE: Severity Levels & Response Times*

Incidents are categorized by severity:

Severity	Definition	Response Time	Resolution Target	Support Hours
Critical	Complete outage, no user access, data loss risk	15 minutes	2 hours	24×7
High	Major feature unavailable, significant impact to key users	2 hours	8 hours	24×7
Medium	Feature partially degraded, workaround available	24 hours	3 days	Business hours (10 AM–6 PM IST)
Low	Minor UI bug, cosmetic issue, no business impact	48 hours	7 days	Business hours (10 AM–6 PM IST)

Critical Incident Process: 1. Vendor on-call engineer notified immediately (SMS + call + Slack) 2. War room conference call initiated within 15 minutes 3. Status updates provided every 15 minutes until resolution 4. Post-incident report (root cause, timeline, preventive actions) within 24 hours of resolution

13.5.4 Quality & Defect Management

13.5.4.1 *CLAUSE: Defect Classification & Warranty*

Defects discovered during UAT and within 90 days of production go-live shall be corrected by Vendor at no additional cost. Defects are classified as:

Defect Severity	Definition	Fix Timeline	Re-Test Timeline
Blocker	Core feature non-functional, prevents usage	1 business day	Next business day
Critical	Major feature broken or severely degraded	3 business days	Within 5 business days
Major	Features have significant issues but workaround exists	5 business days	Within 7 business days
Minor	Cosmetic or low-impact issue	10 business days	Within 14 business days

Regression Rule:

If a code change introduced by Vendor causes a previously working feature to fail, Vendor must revert or fix the change at no additional cost within 1 business day.

Warranty Period: 90 days post go-live. After 90 days, defect fixes are billed as maintenance at agreed rates.

13.5.5 Acceptance & Milestone Closure

13.5.5.1 *CLAUSE: UAT Acceptance Criteria*

Each milestone is considered complete and acceptable only when:

1. All Functional Requirements for that milestone (per SRS traceability matrix) have passed test cases.
2. No Blocker or Critical defects remain open.
3. Test execution summary and defect report are submitted to the Customer.
4. Customer has completed user acceptance testing (UAT) for 5 business days minimum.

Deemed Acceptance:

If the Customer does not provide a written rejection (with documented reasons linked to SRS criteria) within 5 business days of delivery, the milestone is automatically accepted, and payment is triggered.

Non-Acceptance:

If Customer identifies non-conformities, Vendor shall prioritize fixes and resubmit for acceptance within 5

business days. Repeated rejections (>2 per milestone) shall trigger an escalation to the Steering Committee.

13.6 SECURITY & COMPLIANCE CLAUSES

13.6.1 Security Standards & Controls

13.6.1.1 *CLAUSE: Security Baseline Requirements*

Vendor shall implement and maintain the following security controls aligned with WEZU-SRS-001 (Section 4.2):

Data Encryption: - All sensitive data at rest encrypted with AES-256 (or stronger) - All data in transit encrypted with TLS 1.2 (or higher) - HTTPS enforced on all web endpoints (no HTTP allowed) - Certificate pinning implemented in mobile apps to prevent MITM attacks

Authentication & Authorization: - OAuth2 + JWT token-based authentication for APIs - Multi-factor authentication (MFA) optional for admin/dealer users, enforced for privileged accounts - Biometric login support (fingerprint/face recognition) on mobile apps - Session management: auto-logout after 30 minutes of inactivity, logout from all devices option available - Role-based access control (RBAC) with least-privilege principle

Payment Security: - PCI-DSS Level 1 compliance for payment processing via Razorpay - No credit card data stored on Vendor's servers (tokenization only) - 3D Secure (3DS2) authentication enforced for online transactions - Fraud detection rules and velocity checks implemented

Infrastructure Security: - Regular OS and library patching (monthly security updates) - Vulnerability scanning (automated weekly via SAST/DAST tools) - Penetration testing conducted quarterly (results shared with Customer) - DDoS mitigation service deployed - Web Application Firewall (WAF) configured with rate limiting and IP blocking rules

13.6.2 Security Incident Management

13.6.2.1 *CLAUSE: Breach Notification & Response*

In the event of any suspected or confirmed security breach, data breach, unauthorized access, or compliance violation, Vendor shall:

1. **Immediate Notification (within 2 hours):** Notify Customer's security officer via phone, email, and Slack with initial details: nature of breach, data affected, timing, scope of exposure.
2. **Formal Notice (within 24 hours):** Submit a detailed incident report including:
 - Timeline of discovery and detection
 - Affected data (user count, data categories)
 - Root cause analysis
 - Immediate containment actions taken
 - Legal/regulatory obligations triggered (e.g., mandatory reporting timelines)
3. **Investigation & Remediation (within 7 days):** Complete forensic investigation and provide:

- Detailed root-cause analysis
 - Remediation plan with preventive measures
 - Monitoring/logging enhancement recommendations
4. **Post-Incident:** Conduct a joint post-mortem with Customer and implement agreed remediation steps at Vendor's cost if breach was due to Vendor negligence.
-

13.6.3 Compliance Obligations

13.6.3.1 CLAUSE: Regulatory Compliance

Vendor shall ensure the platform operates in compliance with all applicable Indian laws and regulations, including:

- **Goods and Services Tax (GST):** Proper GST-compliant invoicing, reporting, and remittance.
- **Know Your Customer (KYC) Regulations:** Compliance with PMLA/AML requirements, support for Aadhaar/PAN/GST API integrations.
- **Payment Regulations:** Compliance with RBI guidelines for electronic payments, FDR rules for float management (if applicable).
- **Data Protection:** Adherence to data protection and privacy standards equivalent to GDPR principles (consent, purpose limitation, data retention, deletion).
- **Consumer Protection Act & E-commerce Rules:** Compliance with product liability, warranty, and dispute resolution requirements.

Vendor shall maintain documentation of compliance audit results and make them available for Customer review upon request.

13.6.4 Audit & Compliance Verification

13.6.4.1 CLAUSE: Audit Rights

Customer has the right to:

1. **Request Audit Reports:** Vendor shall provide SOC 2 Type II, ISO 27001, or equivalent third-party audit reports within 10 business days of request.
2. **Conduct Security Assessments:** Customer may conduct security questionnaires or brief technical assessments (30–60 minutes) with Vendor's technical team, with reasonable notice (7 days).
3. **Review Security Documentation:** Vendor shall maintain and share architecture diagrams, security policies, access control matrices, and vulnerability scanning results (with Vendor's approval for sensitive details).

Audit Schedule: Vendor shall undergo an independent security audit (SOC 2 or equivalent) at least annually. Results shall be shared with Customer within 30 days of completion.

13.7 INTELLECTUAL PROPERTY CLAUSES

13.7.1 Ownership of Deliverables

13.7.1.1 *CLAUSE: Customer Ownership of Custom Work*

Customers own all intellectual property rights, including copyrights, patents, and trade secrets, in and to:

- All custom source code written specifically for the WEZU platform
- Custom data models, database schemas, and API designs
- Business logic, workflows, and feature implementations
- Custom documentation, architecture diagrams, and technical specifications
- All customizations, configurations, and extensions to third-party software

Vendor hereby assigns all right, title, and interest in such custom IP to the Customer, and shall execute all necessary documents to perfect such assignment at Customer's request (at no additional cost).

13.7.2 Vendor Retained IP & License Grant

13.7.2.1 *CLAUSE: Vendor Reusable Components & Licenses*

Vendor retains ownership of pre-existing, reusable components, frameworks, tools, and methodologies created by Vendor prior to this engagement or used in other customer engagements, including:

- Development frameworks and libraries (React, Node.js, etc.)
- Vendor's proprietary build, testing, deployment automation tools
- Vendor's project management and documentation templates
- Vendor's general methodologies and best practices

License Grant: Vendor hereby grants Customer a worldwide, perpetual, irrevocable, royalty-free license to use such Vendor IP solely as embedded in, or necessary to use, the custom deliverables of this engagement. Customer may not reverse engineer, sublicense, or sell such Vendor IP separately.

13.7.3 Third-Party IP & Open Source

13.7.3.1 *CLAUSE: Third-Party Compliance*

Vendor warrants that:

1. All third-party libraries, frameworks, and tools used in the platform shall comply with industry-standard open-source licenses (MIT, Apache 2.0, BSD, etc.).
2. No copyleft-licensed code (GPL v3, AGPL) shall be included unless:
 - a. Explicitly approved by Customer in writing, OR
 - b. Used only in non-commercial/development environments not deployed to production.
3. Vendor shall maintain a Software Bill of Materials (SBOM) listing all third-party components, versions, licenses, and known vulnerabilities, updated monthly.

4. Vendor shall notify Customer immediately upon discovery of security vulnerabilities in third-party dependencies and provide patches or workarounds within the incident response SLA.
-

13.7.4 IP Indemnity

13.7.4.1 CLAUSE: IP Infringement Indemnity

Vendor shall defend, indemnify, and hold harmless Customer from and against any third-party claims, damages, costs, and legal fees arising out of allegations that the custom deliverables or Vendor's work infringes any third-party intellectual property rights (patents, copyrights, trademarks, trade secrets).

Vendor's Remedies: If an infringement claim arises, Vendor shall, at its sole expense and option: - Obtain the right for Customer to continue using the deliverables, OR - Replace or modify the deliverables to make them non-infringing, OR - If the above remedies are not commercially feasible, terminate the engagement and refund fees paid for the infringing component (prorated).

13.8 DATA PROTECTION & PRIVACY CLAUSES

13.8.1 Data Ownership & Processing

13.8.1.1 CLAUSE: Data Ownership

Customer owns all data processed, stored, or transmitted by the platform, including:

- User profile data, KYC information (Aadhaar, PAN, GST numbers)
- Battery rental and purchase transaction records
- GPS location and IoT telemetry data
- Financial/payment data
- Analytics and derived insights (demand forecasts, fraud scores)

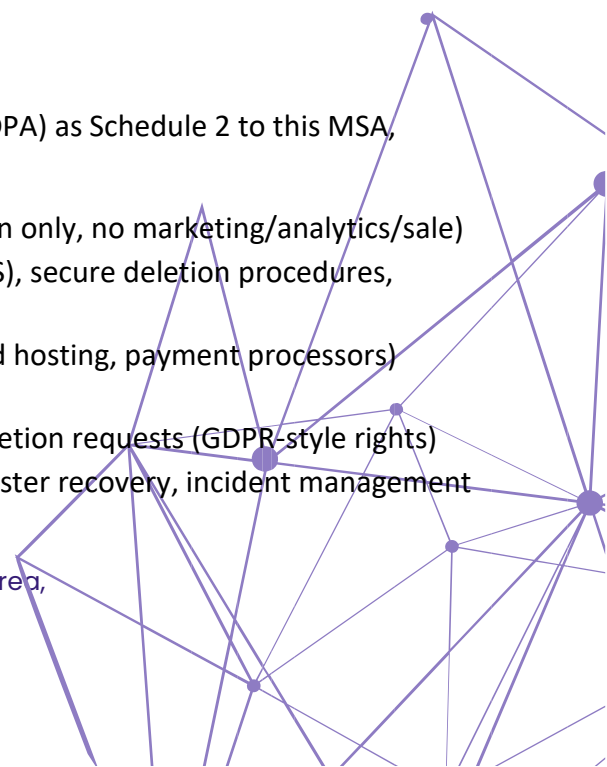
Vendor acts solely as a Data Processor, with no independent rights to use such data except as necessary to provide the services contracted herein.

13.8.2 Data Processing Agreement (DPA)

13.8.2.1 CLAUSE: Data Protection Framework

Vendor and Customer shall execute a Data Processing Agreement (DPA) as Schedule 2 to this MSA, covering:

- **Purposes & Scope:** Permissible uses of data (service provision only, no marketing/analytics/sale)
- **Retention & Deletion:** Data retention periods (aligned to SRS), secure deletion procedures, certification of deletion upon termination
- **Sub-processors:** Any third-party service providers (e.g., cloud hosting, payment processors) processing data on Vendor's behalf
- **Data Subject Rights:** Support for data export, portability, deletion requests (GDPR-style rights)
- **Security Measures:** Encryption, access controls, backup, disaster recovery, incident management



- **Audit & Monitoring:** Customer right to audit data handling practices, including security assessments and compliance questionnaires
- **Cross-Border Transfers:** Confirmation that all data is stored within Indian jurisdiction; any offshore access requires explicit written approval

13.8.3 Confidentiality & non-disclosure

13.8.3.1 *CLAUSE: Confidentiality Obligations*

Each party ("Receiving Party") agrees to:

1. **Protect Confidential Information** of the other party ("Disclosing Party") using at least the same degree of care as it uses for its own confidential information, but not less than reasonable care.
2. **Limit Access** to confidential information to employees, contractors, and advisors on a need-to-know basis who have signed confidentiality agreements.
3. **Permitted Disclosures:** Confidential information may be disclosed only:
 - To the extent required by law (with advance notice to allow Disclosing Party to seek protective orders)
 - To legal/regulatory advisors under attorney-client privilege
 - As necessary for performance of services under this MSA
4. **Return/Destruction:** Upon termination, return or destroy (at Disclosing Party's election) all confidential information except where retention is required by law.

Exceptions:

Information that is (a) publicly available, (b) independently developed, (c) rightfully received from third parties without confidentiality obligations, or (d) general knowledge of the Receiving Party's employees is not confidential.

13.9 EXIT, TRANSITION & TERMINATION CLAUSES

13.9.1 Termination Rights

13.9.1.1 *CLAUSE: Termination for Cause*

Customer may terminate this Agreement immediately upon written notice if:

1. **Material Breach:** Vendor materially breaches obligations under this MSA and fails to cure within 30 days of written notice specifying the breach.
2. **Persistent SLA Failure:** Vendor fails to meet agreed SLAs for three (3) consecutive months, with service credits exhausted.
3. **Security Breach:** A confirmed security breach occurs due to Vendor's negligence, resulting in unauthorized access, data exfiltration, or regulatory fines.
4. **Insolvency:** Vendor becomes insolvent, bankrupt, or is subject to receivership; files for bankruptcy protection; or has creditors' rights suspended.
5. **Compliance Violation:** Vendor violates applicable law or regulatory requirements (GST, KYC, data protection, etc.) in performance of services.

13.9.2 Termination for Convenience

13.9.2.1 *CLAUSE: Early Termination Option*

Customer may terminate this Agreement for convenience with:

- **Phase 1:** 60 days' written notice, with early termination charges of 15% of remaining Phase 1 fees
- **Phase 2:** 30 days' written notice, with early termination charges of 20% of Phase 2 fees already invoiced (actual costs incurred by Vendor)

Upon early termination, Vendor will continue providing services through the notice period and deliver all work-in-progress and documentation.

13.9.3 Transition Assistance & Data Export

13.9.3.1 *CLAUSE: Exit Support Obligations*

Upon expiration or termination of this Agreement, Vendor shall, at no additional cost (except for extended transition support beyond 30 days, which shall be billed at agreed rates):

Data Export: - Within 5 business days: Provide complete data extracts in standard formats (CSV, JSON, SQL dumps) of all customer data, transaction records, IoT telemetry, and configuration settings - Ensure data completeness and integrity (checksums, record counts provided) - Verify data against production database (audit trail provided)

Documentation & Knowledge Transfer: - Within 10 business days: Deliver all technical documentation, including: - Architecture and design documents (with refresh if >3 months old) - API documentation (Swagger/OpenAPI specs) - Database schemas and data dictionaries - Deployment runbooks and operational procedures - Security & compliance documentation - Admin/configuration manuals - Conduct 2–3 knowledge transfer sessions (4–6 hours total) with Customer's technical team to cover platform architecture, maintenance, and troubleshooting

Infrastructure Handover: - Provide AWS account credentials and console access (if hosted on AWS) or assist with infrastructure migration to Customer's own cloud account - Transfer domain registrations, SSL certificates, and third-party API accounts (Razorpay, Google Maps, etc.) to Customer's control - Update DNS records and routing to point to Customer's infrastructure

Extended Transition Support: - Optional up to 90 days beyond termination date at ₹[INSERT]/hour for incident resolution and platform stability support - Vendor shall remain available for critical production incidents (Severity 1) during this period

13.9.4 Data Deletion & Archival

13.9.4.1 *CLAUSE: Secure Data Destruction*

Upon Customer's request (or 90 days after termination, whichever is earlier):

1. Vendor shall securely delete all copies of Customer data from production, backup, and archive systems.
2. Deletion shall be performed using NIST SP 800-88 secure erasure standards (cryptographic wipe or physical destruction).
3. Vendor shall provide a **Certificate of Destruction**, signed by an authorized officer, confirming:
 - Data categories destroyed
 - Deletion date and method
 - Number of copies destroyed (production + backups)
 - Certification that no residual copies exist

Exceptions:

Customer may request retention of certain data (e.g., for regulatory compliance, litigation hold) beyond the standard deletion period. Retention terms and associated costs shall be documented in writing.

13.9.5 Survival Clauses

13.9.5.1 *CLAUSE: Surviving Obligations*

The following clauses shall survive termination or expiration of this Agreement:

- IP Ownership (Clause 5.1)
- Confidentiality (Clause 6.3)
- Limitation of Liability (Clause 8.3)
- Indemnities (Clause 5.4, 8.2)
- Data Protection & Deletion (Clause 7.4)
- Dispute Resolution & Governing Law (Clause 8.4–8.5)
- Any payment obligations for invoices issued before termination date

13.10 RISK ALLOCATION & LIABILITY CLAUSES

13.10.1 Warranties & Representations

13.10.1.1 *CLAUSE: Vendor Warranties*

Vendor warrants that:

1. **Performance:** Services will be performed with reasonable skill, care, and diligence in accordance with industry's best practices.
2. **Conformity to Specifications:** All deliverables shall conform in all material respects to the functional and non-functional requirements in WEZU-SRS-001 (attached as Schedule 1).
3. **Authority & Capacity:** Vendor has full legal authority to enter into this Agreement and deliver the services; no third-party consents are required.
4. **IP Non-Infringement:** The custom deliverables will not infringe any third-party intellectual property rights.
5. **Security & Compliance:** Services shall be provided in compliance with all applicable laws, security standards (as per SRS Section 4.2), and data protection regulations.

6. **Personnel Qualification:** All personnel assigned to the project are qualified, trained, and screened (background check) as appropriate for their roles.

Warranty Period: These warranties apply throughout the engagement and survive for 90 days post go-live for defect-related claims.

13.10.2 Indemnities

13.10.2.1 CLAUSE: Mutual Indemnities

Vendor's Indemnity:

Vendor shall defend, indemnify, and hold harmless Customer from any third-party claims, losses, damages, and legal fees arising from:

- Vendor's breach of this Agreement
- Vendor's infringement of third-party IP rights
- Vendor's negligence or willful misconduct
- Unauthorized access, data breach, or loss of data due to Vendor's security lapses
- Regulatory fines/penalties resulting from Vendor's non-compliance (GST evasion, KYC non-compliance, etc.)

Customer's Indemnity:

Customer shall defend, indemnify, and hold harmless Vendor from claims arising from:

- Customer data, content, or materials provided by Customer that infringe third-party rights
- Customer's use of the platform in violation of law
- Customer's breach of this Agreement

13.10.3 Limitation of Liability

13.10.3.1 CLAUSE: Capped Liability

Except as expressly provided in this Clause, each party's total liability arising out of or related to this Agreement shall not exceed:

CAPPED LIABILITY = (Fees Paid in Preceding 12 Months) × 1.5
OR
₹50,000,000 (Fifty Million Indian Rupees)
whichever is greater

Uncapped Liability (Exceptions):

Notwithstanding the above cap, the following shall be uncapped:

- Intellectual property infringement claims (IP indemnity)
- Data breaches, unauthorized access, loss of confidential data (due to Vendor negligence)
- Regulatory fines/penalties (due to Vendor's non-compliance with law)
- Either party's willful misconduct, fraud, or gross negligence

- Either party's indemnification obligations
-

13.10.4 Exclusion of Consequential Damages

13.10.4.1 CLAUSE: No Consequential Damages

Except for Uncapped Liability events (Clause 8.3.1), **neither party shall be liable for any indirect, incidental, consequential, special, or punitive damages**, including:

- Lost profits, lost revenue, lost business opportunity
- Lost data or data corruption (except where Vendor is directly at fault)
- Business interruption or downtime losses
- Reputational or goodwill damage

This exclusion applies regardless of the cause of action (contract, tort, negligence, etc.) and whether the party was advised of the possibility of such damage.

13.10.5 Mitigation & Insurance

13.10.5.1 CLAUSE: Risk Mitigation

Customer acknowledges that Vendor has limited ability to prevent all security threats and data loss risks. Accordingly, Customer agrees to:

1. Maintain its own data backups and disaster recovery procedures
2. Implement additional security controls where needed for its risk profile
3. Use the platform only for lawful purposes and in accordance with this Agreement

Insurance: Vendor shall maintain the following insurance policies:

- **Cyber Liability Insurance:** Minimum ₹100 Lakhs coverage for data breach, privacy violation
- **Professional Indemnity Insurance:** Minimum ₹50 Lakhs for errors & omissions
- **General Commercial Liability:** Minimum ₹50 Lakhs

Proof of insurance (certificates of coverage) shall be provided to Customer upon request.

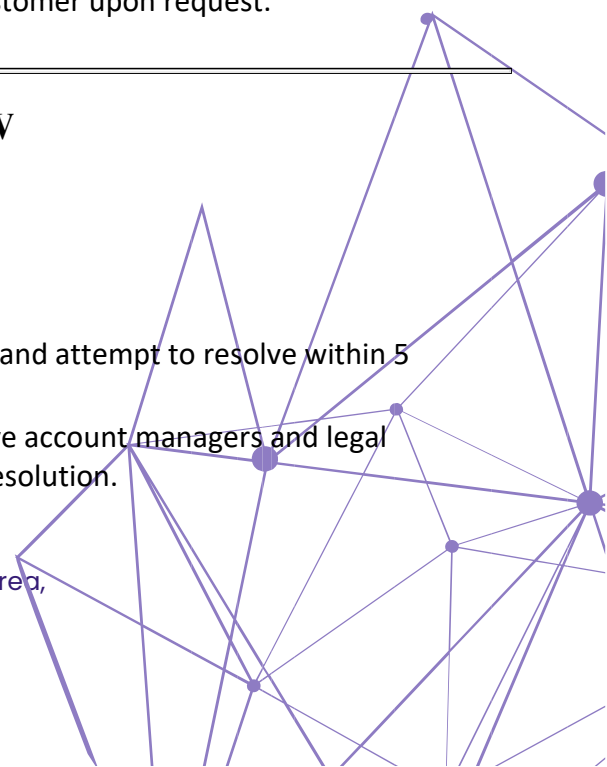
13.11 DISPUTE RESOLUTION & GOVERNING LAW

13.11.1 Escalation & Negotiation

13.11.1.1 CLAUSE: Dispute Resolution Process

In the event of a dispute:

1. **Operational Resolution (Day 1–5):** Project teams shall meet and attempt to resolve within 5 business days.
2. **Management Escalation (Day 6–15):** If unresolved, respective account managers and legal representatives shall meet within 10 business days to seek resolution.



3. **Executive Escalation (Day 16+):** If still unresolved, the matter shall be escalated to the respective executive sponsors (Customer's CFO/CTO, Vendor's MD) for final negotiation and decision.
-

13.11.2 Mediation & Arbitration

13.11.2.1 CLAUSE: Binding Arbitration

If escalation does not resolve the dispute within 45 days:

1. **Mediation:** The parties shall attempt mediation under the Indian Council of Arbitration (ICA) mediation rules, with a mutually agreed mediator.
2. **Arbitration:** If mediation fails, the dispute shall be resolved by **binding arbitration** under the Arbitration and Conciliation Act, 1996 (India):
 - **Venue:** Bangalore, India
 - **Arbitrator(s):** Single arbitrator (or three arbitrators if claim exceeds ₹1 Crore) mutually agreed by the parties
 - **Language:** English
 - **Governing Law:** Laws of India (excluding conflict-of-laws provisions)

Arbitration shall be confidential, and the arbitration award shall be final and binding.

13.11.3 Injunctive Relief

13.11.3.1 CLAUSE: Irreparable Harm

Notwithstanding the arbitration clause, either party may seek injunctive or other equitable relief in a court of competent jurisdiction (Bangalore) to prevent irreparable harm from breach of confidentiality, IP misappropriation, or security violations.

13.12 GENERAL PROVISIONS

13.12.1 Entire Agreement

13.12.1.1 CLAUSE: Integration

This MSA, together with the attached Schedules (including WEZU-SRS-001, SOW, DPA), constitutes the entire agreement between the parties regarding the subject matter and supersedes all prior negotiations, understandings, and agreements (written or oral).

No amendment, modification, or waiver of this Agreement is valid unless in writing and signed by authorized representatives of both parties.



13.12.2 Severability

13.12.2.1 CLAUSE: Severability

If any provision of this Agreement is found to be invalid, illegal, or unenforceable, such provision shall be severed, and the remaining provisions shall continue in full force and effect. If necessary, the severed provision shall be reformed to achieve the intent of the parties.

13.12.3 Force Majeure

13.12.3.1 CLAUSE: Force Majeure Events

Neither party shall be liable for failure to perform any obligation under this Agreement due to causes beyond its reasonable control, including:

- Natural disasters (earthquakes, floods, hurricanes)
- Government action, war, terrorism
- Pandemics or epidemics (e.g., COVID-19)
- Telecommunications/internet infrastructure failures (>4 hours)
- Strikes, labor disputes

Exception: Force majeure does not excuse payment obligations or data security obligations, except to the extent performance is physically impossible.

13.12.4 Notices

13.12.4.1 CLAUSE: Notice Procedures

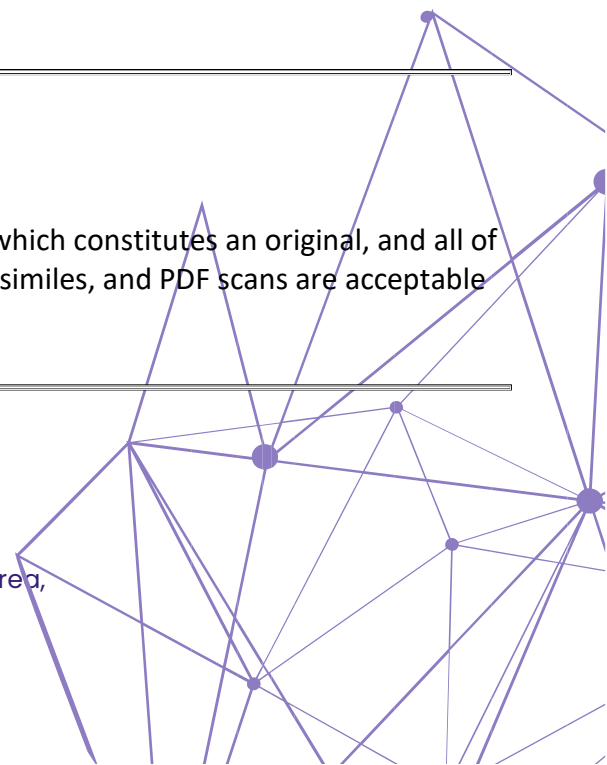
All notices required under this Agreement shall be in writing and delivered:

- **To Customer:** [Customer's Legal Contact Name], [Address], email: [email@domain.com]
 - **To Vendor:** [Vendor's Legal Contact Name], [Address], email: [email@domain.com]
 - **To service provider** [Service Provider's Legal Contact Name], [Address], email: [email@domain.com]
 - Notices are effective upon receipt (email delivery is acceptable, with read receipt or response confirming receipt).
-

13.12.5 Counterparts & Digital Signatures

13.12.5.1 CLAUSE: Execution

This Agreement may be executed in multiple counterparts, each of which constitutes an original, and all of which together constitute one instrument. Electronic signatures, facsimiles, and PDF scans are acceptable and have the same legal effect as original signatures.



13.13 SCHEDULES

13.13.1 Schedule 1: WEZU-SRS-001 (Energy Monetization Platform Software Requirements Specification)

- [Attached separately]
- Defines all Functional Requirements (FR-MOB-, FR-ADMIN-, etc.) and Non-Functional Requirements (NFR-*)
- Acceptance Criteria and test cases referenced throughout this Clause Library

13.13.2 Schedule 2: Statement of Work (SOW) – Phase 1

- Project timeline, phases, milestones, deliverables
- Resource allocation
- Pricing and payment schedule

13.13.3 Schedule 3: Data Processing Agreement (DPA)

- Data ownership, processing scope, retention policies
- Sub-processor management
- Data subject rights (export, deletion)
- Security & breach notification procedures

13.13.4 Schedule 4: Service Level Agreement (SLA) – Detailed Metrics

- Uptime measurement methodology
- Performance KPI baseline tests
- Incident response procedures
- Service credit calculation

13.13.5 Schedule 5: Fee Schedule & Rate Card

- Phase 1 milestone fees
- T&M rates for CRs (Senior Dev, Dev, QA, DevOps)
- Phase 2 optional engagement rates
- Transition support rates (₹/hour)

13.14 SIGNATURE PAGE

MASTER SERVICES AGREEMENT – WEZU ENERGY MONETIZATION PLATFORM

IN WITNESS WHEREOF, the parties have executed this Agreement as of the date first written above.

CUSTOMER:

[Customer Company Name]

By: _____

Name: [Name]

Title: [Title]

Date: _____

Authorized on behalf of Customer

VENDOR:

[Vendor Company Name]

By: _____

Name: [Name]

Title: [Title]

Date: _____

Authorized on behalf of Vendor

SERVICE PROVIDER:

[Service Provider Company Name]

By: _____

Name: [Name]

Title: [Title]

Date: _____

Authorized on behalf of Service provider

